Electromagnetic Waves

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Wave Equations

1. **The wave equation in one dimension:**

$$\frac{\partial^2 f(t,x)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 f(t,x)}{\partial t^2}$$

2. **The wave equation in three-dimensional electromagnetic field:**

$$\begin{cases} \nabla^2 \mathbf{E} = \dfrac{1}{v^2}\dfrac{\partial^2}{\partial t^2}\mathbf{E} \\ \\ \nabla^2 \mathbf{B} = \dfrac{1}{v^2}\dfrac{\partial^2}{\partial t^2}\mathbf{B} \end{cases} \qquad v = \frac{1}{\sqrt{\mu\epsilon}}.$$

In vacuum, $v = c = 1/\sqrt{\mu_0\epsilon_0}$.

3. **General solution to the electric plane waves:**

$$\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\left[\int \mathrm{d}^3\mathbf{k}\,\tilde{\mathbf{E}}(\mathbf{k})e^{i\left(\mathbf{k}\cdot\mathbf{r}-\omega(\mathbf{k})t\right)}\right]$$

4. **The delta function identity:**

$$\int_{-\infty}^{\infty} \mathrm{d}k\, e^{ikx} = 2\pi\delta(x)$$

5. **Fourier Transform in N-dimension :**

$$\begin{cases} \mathbf{E}(\mathbf{r}) = \displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}^N\mathbf{k}}{\mathcal{N}_1}\tilde{\mathbf{E}}(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r})} \\ \\ \tilde{\mathbf{E}}(\mathbf{k}) = \displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}^N\mathbf{r}}{\mathcal{N}_2}\mathbf{E}(\mathbf{r})e^{-i(\mathbf{k}\cdot\mathbf{r})} \end{cases} \qquad \mathcal{N}_1\mathcal{N}_2 = (2\pi)^N$$

1 Fourier Transform

1.1 Delta Function Identity

Consider a function which will diverge from $-\infty \to \infty$:

$$I = \int_{-\infty}^{\infty} e^{ik(x-x')}\,\mathrm{d}k \to \infty. \tag{1}$$

In general, the integral is only valid in the sense of distribution. But first, we need to consider the integral:

$$I_\epsilon(x) = \lim_{\epsilon\to 0}\int_{-\infty}^{\infty} \mathrm{d}k\, e^{ik(x-x')}e^{i\epsilon k^2}. \tag{2}$$

Consider the identity of Gaussian integral[1]:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2+bx} = \sqrt{\frac{\pi}{a}}e^{b^2/4a}, \tag{3}$$

where $a = \epsilon$ and $b = i(x - x')$ in (2). Therefore:

$$I_\epsilon = \lim_{\epsilon\to 0}\sqrt{\frac{\pi}{\epsilon}}\exp\left[-\frac{(x-x')^2}{4\epsilon}\right] = \lim_{\epsilon\to 0}\sqrt{\frac{\pi}{\epsilon}}\frac{1}{\exp\left[(x-x')^2/4\epsilon\right]}. \tag{4}$$

Note that there must be some problems with the result of the integral I_ϵ:

- If $x = x'$, then the exponential term will be 1. Since $\epsilon \to 0$, so the term $\sqrt{\pi/\epsilon}$ will diverge.
- If $x \neq x'$, then the exponential term will decay to zero since the decay speed of exponential functions is far faster than the polynomials.

In general, we have a well-defined test function $\phi(x)$, the distribution says

$$\langle I_\epsilon, \phi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, \lim_{\epsilon\to 0}\sqrt{\frac{\pi}{\epsilon}}\exp\left[-\frac{(x-x')^2}{4\epsilon}\right]\phi(x) = \lim_{\epsilon\to 0}\sqrt{\frac{\pi}{\epsilon}}\int_{-\infty}^{\infty}\mathrm{d}x\, e^{-\left(\frac{x-x'}{2\sqrt{\epsilon}}\right)^2}\phi(x)$$

Let $u = (x - x')/2\sqrt{\epsilon}$, then $2\sqrt{\epsilon}du = \mathrm{d}x$, and $x = x' + 2u\sqrt{\epsilon}$. Therefore, we obtain:

$$\begin{aligned}\langle I_\epsilon, \phi\rangle_{\epsilon\to 0} &= \lim_{\epsilon\to 0}\sqrt{\frac{\pi}{\epsilon}}\int_{-\infty}^{\infty} 2\sqrt{\epsilon}\, du\, e^{-u^2} f(x' + 2u\sqrt{\epsilon}) \\ &= 2\sqrt{\pi}\lim_{\epsilon\to 0}\int_{-\infty}^{\infty} du\, e^{-u^2} f(x' + 2u\sqrt{\epsilon}) = 2\sqrt{\pi} f(x')\underbrace{\int_{-\infty}^{\infty} du\, e^{-u^2}}_{=\sqrt{\pi}} = 2\pi f(x').\end{aligned}$$

[1] See *Mathematical Physics*.

After the argument and evaluation, the integral I at the beginning has the meaning under the distribution. The result above shows that the distribution is equivalent to:

$$\langle I_\epsilon, \phi\rangle_{\epsilon\to 0} = \langle I, \phi\rangle = \langle 2\pi\delta, \phi\rangle = \phi(x'),$$

i.e.

$$\boxed{I = \int_{-\infty}^{\infty} \mathrm{d}k\, e^{ik(x-x')} = 2\pi\delta(x-x').} \tag{5}$$

In 3-dimension, the integrating element k becomes $\mathbf{k} = (k_x, k_y, k_z)$, and the domain D_x becomes $D_\mathbf{r}$, where $\mathbf{r} = (x, y, z)$.[2] The integral is written as

$$I = \int_{\mathbb{R}^3} \mathrm{d}^3\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r})} = \int \mathrm{d}^3\mathbf{k}\, e^{i(k_x x + k_y y + k_z z)}, \tag{6}$$

which is equal to

$$I = \int_{D_x} \mathrm{d}x\, e^{ik_x x} \int_{D_y} \mathrm{d}y\, e^{ik_y y} \int_{D_z} \mathrm{d}z\, e^{ik_z z} = (2\pi)^3\delta(x)\delta(y)\delta(z) = (2\pi)^3\delta^3(\mathbf{r}). \tag{7}$$

More general, when $\mathbf{r} \to \boldsymbol{\mathscr{r}}$, the integral becomes

$$I = \int \mathrm{d}^3\mathbf{k}\, e^{i\mathbf{k}\cdot\boldsymbol{\mathscr{r}}} = 8\pi^3\delta(x-x')\delta(y-y')\delta(z-z') = 8\pi^3\delta^3(\boldsymbol{\mathscr{r}}). \tag{8}$$

1.2 Fourier Transform and Representation

A well-defined(sometimes not, but I don't really care) function $f(x)$ can be denoted by an infinite periodic function[3] with respect to the frequency k, this is called *Fourier Transformation*. The Fourier transform is defined as

$$f(x) = \int_{-\infty}^{\infty} \frac{\mathrm{d}k}{\mathcal{N}_1} \tilde{f}(k) e^{ikx}, \quad \tilde{f}(k) = \int_{-\infty}^{\infty} \frac{\mathrm{d}x'}{\mathcal{N}_2} f(x') e^{-ikx'}. \tag{9}$$

WLOG, substituting $\tilde{f}(k)$ into the $f(x)$ transformation, we must have

$$f(x) = \int_{-\infty}^{\infty} \frac{\mathrm{d}k}{\mathcal{N}_1} \left[\int_{-\infty}^{\infty} \frac{\mathrm{d}x'}{\mathcal{N}_2} f(x') e^{-ikx'}\right] e^{ikx}. \tag{10}$$

[2] From this argument, we can predict that this new integral can be considered to be the distribution under three bases, or directions.

[3] Continuous Fourier series

After the substitution, the equation must hold. Then we expand the integral:

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{\mathrm{d}k}{\mathcal{N}_1}\left[\int_{-\infty}^{\infty} \frac{\mathrm{d}x'}{\mathcal{N}_2} f(x') e^{-ikx'}\right] e^{ikx} &= \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int_{-\infty}^{\infty} \mathrm{d}k \int_{-\infty}^{\infty} \mathrm{d}x' f(x') e^{ik(x-x')} \\
&= \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int_{-\infty}^{\infty} \mathrm{d}x' f(x') \int_{-\infty}^{\infty} \mathrm{d}k\, e^{ik(x-x')} \\
&= \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int_{-\infty}^{\infty} \mathrm{d}x' f(x') \cdot 2\pi\delta(x-x'),
\end{aligned}$$

and the integral is $f(x)$, i.e.

$$f(x) = \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int_{-\infty}^{\infty} \mathrm{d}x' f(x') \cdot 2\pi\delta(x-x') = \frac{2\pi}{\mathcal{N}_1\mathcal{N}_2} f(x). \tag{11}$$

To make the equation hold, it's obvious that

$$\boxed{\mathcal{N}_1\mathcal{N}_2 = 2\pi} \quad \text{1D.} \tag{12}$$

I'm going to generalize the result to 3-D. Consider a function of the position vector $f(\mathbf{r})$ can also be applied the Fourier transform:

$$f(\mathbf{r}) = \int \frac{\mathrm{d}^3\mathbf{k}}{\mathcal{N}_1} \tilde{f}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{r})}, \quad \tilde{f}(\mathbf{k}) = \int \frac{\mathrm{d}^3\mathbf{r}'}{\mathcal{N}_2} f(\mathbf{r}') e^{-i(\mathbf{k}\cdot\mathbf{r}')}. \tag{13}$$

Similarly,

$$\begin{aligned}
f(\mathbf{r}) &= \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int \mathrm{d}^3\mathbf{k}\left[\int \mathrm{d}^3\mathbf{r}' f(\mathbf{r}') e^{-i(\mathbf{k}\cdot\mathbf{r}')}\right] e^{i(\mathbf{k}\cdot\mathbf{r})} \\
&= \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int \mathrm{d}^3\mathbf{r}' f(\mathbf{r}') \int \mathrm{d}^3\mathbf{k}\, e^{i\left[\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')\right]},
\end{aligned}$$

by (8), we obtain

$$f(\mathbf{r}) = \frac{1}{\mathcal{N}_1\mathcal{N}_2}\int \mathrm{d}^3\mathbf{r}' f(\mathbf{r}')(2\pi)^3\delta(\boldsymbol{\mathscr{r}}) = \frac{(2\pi)^3}{\mathcal{N}_1\mathcal{N}_2}\int \mathrm{d}^3\mathbf{r}' f(\mathbf{r}')\delta(\boldsymbol{\mathscr{r}}). \tag{14}$$

Therefore,

$$\boxed{\mathcal{N}_1\mathcal{N}_2 = (2\pi)^3} \quad \text{3D.} \tag{15}$$

Again, this can be generalized to N-dimensions. This transformation is also valid if the function becomes a vector function in N-dimensions:

$$\mathbf{F}(\mathbf{r}) = \frac{1}{\mathcal{N}_1}\int \mathrm{d}^N\mathbf{k}\, \tilde{\mathbf{F}}(\mathbf{k})\, e^{i(\mathbf{k}\cdot\mathbf{r})}, \quad \tilde{\mathbf{F}}(\mathbf{k}) = \frac{1}{\mathcal{N}_2}\int \mathrm{d}^N\mathbf{r}'\, \mathbf{F}(\mathbf{r}')\, e^{-i(\mathbf{k}\cdot\mathbf{r})}. \tag{16}$$

Since the exponential term is Euler's formula $e^{ix} = \cos x + i\sin x$, so it's ok to add some phase difference into the transformation without the validity of it, for example,

$$\mathbf{F}(\mathbf{r}) = \frac{1}{\mathcal{N}_1}\int \mathrm{d}^N\mathbf{k}\, \tilde{\mathbf{F}}(\mathbf{k})\, e^{i(\mathbf{k}\cdot\mathbf{r})} \to \mathbf{F}(\mathbf{r}) = \frac{1}{\mathcal{N}_1}\int \mathrm{d}^N\mathbf{k}\, \tilde{\mathbf{F}}(\mathbf{k})\, e^{i\left(\mathbf{k}\cdot\mathbf{r}\pm\omega(\mathbf{k})t\right)}. \tag{17}$$

2 General Solution to Wave Functions

2.1 Superposition of Waves

Consider a wave function $\phi(x, t)$ that satisfies the wave equation:

$$\frac{\partial^2 \phi(x,t)}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \phi(x,t)}{\partial t^2}. \tag{18}$$

This can also be rewritten as

$$\left(\frac{\partial^2}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\phi(x,t) = \left(\frac{\partial}{\partial x} - \frac{1}{c}\frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial x} + \frac{1}{c}\frac{\partial}{\partial t}\right)\phi = 0, \tag{19}$$

where we define

$$\left(\frac{\partial}{\partial x} + \frac{1}{c}\frac{\partial}{\partial t}\right)\phi(x,t) = \Psi(x,t). \tag{20}$$

We use the change of variables $x_\pm = x \pm ct$, or say, 'let'. Then we can write the origin variables in terms of x_+ and x_-:

$$\begin{cases} x_+ = x + ct \\ x_- = x - ct \end{cases} \Rightarrow \begin{cases} x = \dfrac{x_+ + x_-}{2} \\ t = \dfrac{x_+ - x_-}{2c} \end{cases}$$

therefore we obtain

$$\frac{\partial}{\partial x_\pm} = \frac{\partial x}{\partial x_\pm}\frac{\partial}{\partial x} + \frac{\partial t}{\partial x_\pm}\frac{\partial}{\partial t} = \frac{1}{2}\left(\frac{\partial}{\partial x} \pm \frac{1}{c}\frac{\partial}{\partial t}\right),$$

since x, t are the function of $x_\pm$. In this situation, the function $\Psi(x,t)$ can be expressed as the function of x_+, x_- naturally, that is, $\Psi(x_+, x_-)$. Substituting (20) into (19), we get

$$\left(\frac{\partial}{\partial x} - \frac{1}{c}\frac{\partial}{\partial t}\right)\Psi(x_+, x_-) = 2\frac{\partial}{\partial x_-}\Psi(x_+, x_-) = 0. \tag{21}$$

This differential equation indicates that the function Ψ is independent of x_-, so

$$\underbrace{\left(\frac{\partial}{\partial x} + \frac{1}{c}\frac{\partial}{\partial t}\right)}_{2\partial x_+}\phi(x_+, x_-) = \Psi(x_+) \Rightarrow 2\frac{\partial}{\partial x_+}\phi(x_+, x_-) - \Psi(x_+) = 0, \tag{22}$$

it's trivial that ϕ is also independent of x_-, so one of the solutions to ϕ can be written as $\phi_+(x_+)$. We can also let

$$\left(\frac{\partial}{\partial x} - \frac{1}{c}\frac{\partial}{\partial t}\right)\phi(x_+, x_-) = \Psi_2(x_+, x_-),$$

which gives another solution to $\phi = \phi_-(x_-)$. Therefore, the general solution to the wave function can always be denoted by

$$\phi(x,t) = \phi_+(x_+) + \phi_-(x_-). \tag{23}$$

Check:

$$4\partial_{x_+}\partial_{x_-}\Big[\phi_+(x_+) + \phi_-(x_-)\Big] = 4\partial_{x_+}\Big[\partial_{x_-}\phi_+(x_+)\Big] + 4\partial_{x_-}\Big[\partial_{x_+}\phi_-(x_-)\Big] = 0 \quad \checkmark$$

The solution indicates that a wave can be represented as the superposition of other waves.

2.2 Complex Solution in 1D

The most familiar wave function is sinusoidal wave, the following argument shows why. Since we have the PDE (19), we can use separation of variables, which means that let

$$\phi(x,t) = \mathrm{X}(x)\mathrm{T}(t),$$

then substituting (2.2) into (19), we have

$$\underbrace{\frac{\mathrm{X}''(x)}{\mathrm{X}}}_{-k^2} - \frac{1}{c^2}\underbrace{\frac{\mathrm{T}''(t)}{\mathrm{T}}}_{-\omega^2} = 0 \Rightarrow \begin{cases} \mathrm{X}(x) = A\cos(kx) + B\sin(kx) \\ \\ \mathrm{T}(t) = A'\cos(\omega t) + B'\sin(\omega t), \end{cases}$$

where $\omega^2 = c^2k^2$, and $\omega = \pm ct$, By Euler's formula,

$$\begin{aligned} A\cos(kx) + B\sin(kx) &= \sqrt{A^2+B^2}(\cos kx\cos\delta + \sin kx \sin\delta) \\ &\equiv A\cos(kx-\delta) = \mathrm{Re}\Big[Ae^{i(kx-\delta)}\Big] \\ &= \mathrm{Re}\Big[\big(Ae^{i\delta}\big)e^{ikx}\Big] \equiv \mathrm{Re}\Big[\tilde{A}_1 e^{ikx}\Big] \; (\tilde{A}_1 \in \mathbb{C}), \end{aligned} \tag{24}$$

similarly

$$A'\cos(\omega t) + B'\sin(\omega t) = \mathrm{Re}\Big[\tilde{A}_2 e^{-i\omega t}\Big]. \tag{25}$$

Therefore, the general solution with the complex number can be denoted by the complex form:

$$\phi(x,t) = \mathrm{Re}\left[\sum_k \tilde{A}e^{i(kx-\omega t)}\right] \to \mathrm{Re}\left[\int \mathrm{d}k\, \tilde{A}e^{i(kx-\omega t)}\right] \quad \text{Continuous}, \tag{26}$$

since by the general solution to the wave function, we know that the wave can be superposed by waves. In the 1D case, there are just two directions of propagation, left or right.

On the other hand, the Fourier transform tells us that the wave solution can be denoted by:

$$\tilde{\phi}_+(x_+) = \tilde{\phi}_+(x+ct) = \int \mathrm{d}k\, \tilde{A}_1(k) e^{ik(x+ct)} = \int \mathrm{d}k\, \tilde{A}_1(k) e^{i(kx-\omega t)}. \tag{27}$$

Note that $\omega = \pm ck$, and meanwhile,

$$\tilde{\phi}_-(x_-) = \tilde{\phi}_-(x-ct) = \int \mathrm{d}k\, \tilde{A}_2(k) e^{ik(x-ct)} = \int \mathrm{d}k\, \tilde{A}_2(k) e^{i(kx-\omega t)}, \tag{28}$$

by (27) and (28), the general solution is

$$\tilde{\phi}(x,t) = \tilde{\phi}_+(x_+) + \tilde{\phi}_-(x_-) = \int \mathrm{d}k\, \tilde{A}(k) e^{i(kx-\omega t)}. \tag{29}$$

Same as (26), but we still need to remember to take the real part when considering the real physical waves.

2.3 Solution in 3D and the Wave Vector

When we generalize the solution to 3D waves, it's more complicated but still has a good version of the superposition of waves. This time, the wave function becomes $\phi(x,y,z,t)$, and can also be denoted by $\phi(\mathbf{r},t)$ where $\mathbf{r} = (x,y,z)$ is a position vector. By separation, let

$$\phi = \mathrm{X}(x)\mathrm{Y}(y)\mathrm{Z}(z)\mathrm{T}(t), \tag{30}$$

it leads to

$$\underbrace{\frac{\mathrm{X}''(x)}{\mathrm{X}}}_{-k_x^2} + \underbrace{\frac{\mathrm{Y}''(y)}{\mathrm{Y}}}_{-k_y^2} + \underbrace{\frac{\mathrm{Z}''(z)}{\mathrm{Z}}}_{-k_z^2} - \frac{1}{c^2}\underbrace{\frac{\mathrm{T}''(t)}{\mathrm{T}}}_{-\omega^2} = 0. \tag{31}$$

Then the solution to the wave function is the linear combination of each direction and time:

$$\phi(\mathbf{r},t) = \begin{pmatrix} \sin(k_x x) \\ \cos(k_x x) \end{pmatrix} \begin{pmatrix} \sin(k_y y) \\ \cos(k_y y) \end{pmatrix} \begin{pmatrix} \sin(k_z z) \\ \cos(k_z z) \end{pmatrix} \begin{pmatrix} \sin(\omega t) \\ \cos(\omega t) \end{pmatrix}. \tag{32}$$

We define a new quantity $\mathbf{k} = (k_x, k_y, k_z)$, which is called **wave vector**[4]. In this case, ω also depends on k, which is

$$\omega^2(k) = c^2\left(k_x^2 + k_y^2 + k_z^2\right) \Rightarrow \omega = \pm c|\mathbf{k}| = \pm ck. \tag{33}$$

Each wave vector $\mathbf{k}$ represents a different mode of propagation. For the wave vector $\mathbf{k}_i = (k_{x_i}, k_{y_i}, k_{z_i})$, the solution is:

$$\tilde{\phi}(\mathbf{r},t) = \tilde{A}_i(\mathbf{k})\left(e^{ik_{x_i}x}\right)\left(e^{ik_{y_i}y}\right)\left(e^{ik_{z_i}z}\right)\left(e^{-i\omega t}\right) = \tilde{A}_i(\mathbf{k}) e^{i(k_{x_i}x + k_{y_i}y + k_{z_i}z - \omega t)},$$

[4] In 1D, it's called wave number, which represents how many phases the wave will propagate when changing the position x.

where $k_{x_i}x + k_{y_i}y + k_{z_i}z$ can be denoted by:

$$k_{x_i}x + k_{y_i}y + k_{z_i}z = \mathbf{k}_i \cdot \mathbf{r}, \tag{34}$$

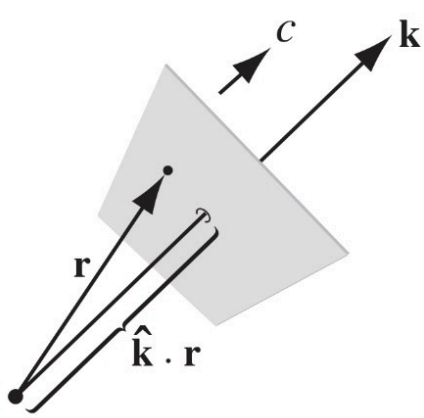
so it can be rewritten as

$$\tilde{\phi}(\mathbf{r}, t) = \tilde{A}_i(\mathbf{k})e^{i(\mathbf{k}_i \cdot \mathbf{r} - \omega t)}. \tag{35}$$

This means that the wave propagates along the direction $\hat{\mathbf{k}}$ since if

$$\mathbf{k}_i \cdot \mathbf{r} - \omega t = \text{const} \Rightarrow \mathbf{r}(t) = \mathbf{r}_0 + \frac{\omega}{k_i^2}\mathbf{k}_i t = \mathbf{r}_0 + \frac{\omega}{k_i}t\hat{\mathbf{k}}_i.$$

In addition, the form of (34) is a plane function in $\mathbb{R}^3$. Therefore, the expression $\mathbf{k} \cdot \mathbf{r} - \omega t$ represents the **phase information** of a wave. This implies that at any given instant in time t, the set of all position vectors $\mathbf{r}$ whose projection onto the direction of propagation ($\mathbf{k}$) results in a constant value defines a geometric plane. On this specific plane, at time t, the phase of the wave is uniform (identical). As shown below:



Because the surfaces of constant phase are parallel planes, this method of three-dimensional propagation is known as a **plane wave**. We've known that waves can be represented as the superposition of all the continuous waves in 3D space along each direction of propagation, hence the general solution to the 3D wave function is:

$$\tilde{\phi}(\mathbf{r}, t) = \sum_{\mathbf{k}} \tilde{A}(\mathbf{k})e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad \Rightarrow \quad \tilde{\phi}(\mathbf{r}, t) = \int \mathrm{d}\mathbf{k}\, \tilde{A}(\mathbf{k})e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}. \tag{36}$$

3 Electromagnetic Problem

3.1 General Solution and Propagation

We consider the environment to be a vacuum, where we want to discuss how the source-free fields generate waves. Since it's source-free, Maxwell's equations become

$$\begin{cases} \boldsymbol{\nabla} \cdot \mathbf{E} = 0 \\ \boldsymbol{\nabla} \cdot \mathbf{B} = 0 \\ \boldsymbol{\nabla} \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \epsilon_0 \dfrac{\partial \mathbf{E}}{\partial t} \end{cases} \tag{37}$$

First, the curl of $\boldsymbol{\nabla} \times \mathbf{E}$ is

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{E}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = \boldsymbol{\nabla} \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \mathbf{B}). \tag{38}$$

Since the divergence of the electric field is 0, the term $\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{E})$ vanishes. Substituting the curl of the magnetic field into (38), then (38) becomes:

$$-\nabla^2 \mathbf{E} = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right), \tag{39}$$

this yields

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{40}$$

Similarly,

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{B}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{B}) - \nabla^2 \mathbf{B} = -\nabla^2 \mathbf{B}$$

$$= \boldsymbol{\nabla} \times \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\boldsymbol{\nabla} \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2},$$

and can also obtain

$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}. \tag{41}$$

(40) and (41) are the *wave equations of the electromagnetic fields.*

$$\begin{cases} \nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \dfrac{\partial^2 \mathbf{E}}{\partial t^2} \\ \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \dfrac{\partial^2 \mathbf{B}}{\partial t^2}. \end{cases} \tag{42}$$

By the property of the wave equation, (42) gives the idea that the electromagnetic field can propagate, with the speed of light c, which satisfies

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.} \tag{43}$$

The general solution to the electric wave equation can be denoted by the linear combination of each direction $\hat{\mathbf{k}}$, which is similar to (36) in the continuous space:

$$\mathbf{E}(\mathbf{r}, t) = \mathrm{Re}\left[\int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{E}}(\mathbf{k}) e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega(\mathbf{k})t\right)}\right], \tag{44}$$

similarly, the general solution to the magnetic wave can also be denoted by:

$$\mathbf{B}(\mathbf{r}, t) = \mathrm{Re}\left[\int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{B}}(\mathbf{k}) e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega(\mathbf{k})t\right)}\right]. \tag{45}$$

I'll use the complex vector to represent the wave since the direction is the most important thing in this chapter. By Gauss's Law in source-free electric field $\boldsymbol{\nabla}\cdot\mathbf{E} = 0$, (44) is applied the divergence to be:

$$\begin{aligned}
\boldsymbol{\nabla}\cdot\mathbf{E} = \boldsymbol{\nabla}\cdot\int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{E}}(\mathbf{k}) e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega(\mathbf{k})t\right)} &= \int \mathrm{d}^3\mathbf{k}\, \boldsymbol{\nabla}\cdot\left[\tilde{\mathbf{E}} e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega t\right)}\right] \\
&= \int \mathrm{d}^3\mathbf{k}\, (\boldsymbol{\nabla}\cdot\tilde{\mathbf{E}}) e^{(\cdots)} + \int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{E}}\cdot\boldsymbol{\nabla} e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega t\right)} \\
&= \int \mathrm{d}^3\mathbf{k}\, i(\tilde{\mathbf{E}}\cdot\mathbf{k}) e^{(\cdots)} = 0,
\end{aligned}$$

Therefore, we can find the important conclusion:

$$\mathbf{k}\cdot\tilde{\mathbf{E}} = 0 \equiv \hat{\mathbf{k}}\cdot\mathbf{E} = 0, \quad \forall\, \hat{\mathbf{k}}_i. \tag{46}$$

So, the electric field is **perpendicular** to the direction of the electric wave which propagates along $\hat{\mathbf{k}}$. Similarly, the curl of the electric field is

$$\begin{aligned}
\boldsymbol{\nabla}\times\mathbf{E} = \boldsymbol{\nabla}\times\int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{E}}(\mathbf{k}) e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega(\mathbf{k})t\right)} &= \int \mathrm{d}^3\mathbf{k}\, \boldsymbol{\nabla}\times\left[\tilde{\mathbf{E}} e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega t\right)}\right] \\
&= \underbrace{\int \mathrm{d}^3\mathbf{k}\, (\boldsymbol{\nabla}\times\tilde{\mathbf{E}}) e^{(\cdots)}}_{=0\,(\tilde{\mathbf{E}} \text{ is } \mathbf{k} \text{ dependence})} - \int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{E}}\times\boldsymbol{\nabla} e^{i\left(\mathbf{k}\cdot\mathbf{r} - \omega t\right)} \\
&= -\int \mathrm{d}^3\mathbf{k}\, i(\tilde{\mathbf{E}}\times\mathbf{k}) e^{(\cdots)} = \int \mathrm{d}^3\mathbf{k}\, {\color{blue} ik(\hat{\mathbf{k}}\times\tilde{\mathbf{E}})} e^{(\cdots)},
\end{aligned}$$

by Faraday's law, we have

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\int \mathrm{d}^3\mathbf{k}\, \tilde{\mathbf{B}} \left[\frac{\partial}{\partial t} e^{i\left(\mathbf{k}\cdot\mathbf{r}-\omega t\right)}\right] = \int \mathrm{d}^3\mathbf{k}\, i\omega\tilde{\mathbf{B}} e^{(\cdots)}.$$

By comparing the results of curl and Faraday's law, we finally get:

$$\omega\tilde{\mathbf{B}} = k(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}) \Rightarrow \tilde{\mathbf{B}} = \frac{k}{\omega}(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}) = \frac{1}{c}(\hat{\mathbf{k}} \times \tilde{\mathbf{E}}). \tag{47}$$

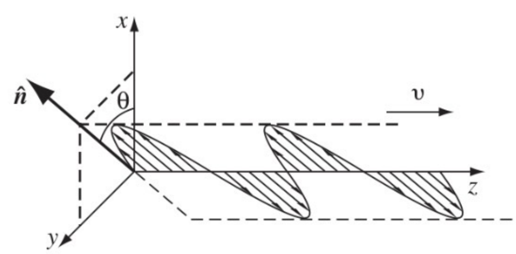
When we take a real part, we can conclude that

$$\begin{cases} \hat{\mathbf{k}} \cdot \mathbf{E}(\mathbf{r}, t) = 0 & \mathbf{E} \perp \hat{\mathbf{k}} \\ \\ \mathbf{B}(\mathbf{r}, t) = \dfrac{1}{c}(\hat{\mathbf{k}} \times \mathbf{E}) & \mathbf{B} \perp \mathbf{E} \perp \hat{\mathbf{k}}. \end{cases} \tag{48}$$

Therefore, once an electric field is given, the corresponding magnetic field is also be determined by the relation (47).

3.2 Polarization

If we choose $z\hat{\mathbf{z}}$ as the direction of propagation, then the electric field will the electric field oscillates along $\hat{\mathbf{n}}$ which is called **Polarization vector** and Due to the transverse nature of electromagnetic waves, the polarization vector must be orthogonal to the direction of propagation:



The polarization vector satisfies

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{z}} = 0, \tag{49}$$

and the direction of the polarization rotates around the $z-$axis in the xy plane. The polarization vector of the wave shown above can be denoted by

$$\hat{\mathbf{n}} = \hat{\mathbf{x}} \cos\theta + \hat{\mathbf{y}} \sin\theta. \tag{50}$$

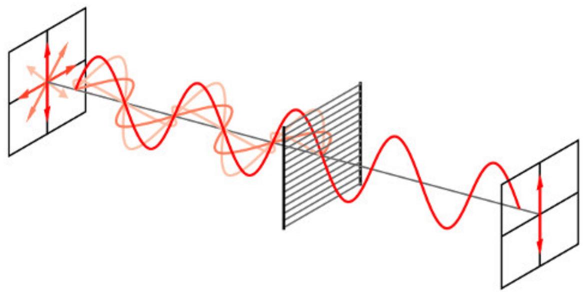
That is, in this case, the electric field is

$$\tilde{\mathbf{E}}_{\text{Prop}} = \tilde{\mathbf{E}}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)} = \hat{\mathbf{z}} \tilde{E}_0 e^{i(kz-\omega t)},$$

since $\mathbf{k} \cdot \mathbf{r} = (0, 0, k) \cdot (x, y, z)$. By (50), the field on the xy plane is

$$\tilde{\mathbf{E}}_{\text{Polar}} = \hat{\mathbf{n}}\tilde{E}_0 e^{i(kz-\omega t)} = \hat{\mathbf{x}}\tilde{E}_0 e^{i(kz-\omega t)} \cos\theta + \hat{\mathbf{y}}\tilde{E}_0 e^{i(kz-\omega t)} \sin\theta. \tag{51}$$

Once we mention the polarization, we must introduce a cool object - a **polarizer**. As shown below:



When the electric fields which propagate along the $z-$axis with different polarization vectors, denoted by $\hat{\mathbf{n}}_i = \hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{y}}\sin\theta_i$ $(i = 1, 2, ...)$ pass through a linear polarizer characterized by a transmission axis $\hat{\mathbf{p}}$ allows only the component of the electric field parallel to $\hat{\mathbf{p}}$ to pass. Components perpendicular to this axis are absorbed or reflected by the material.
This process relies on the principle of superposition. Any arbitrarily polarized plane wave can be decomposed into a linear combination of two orthogonal basis vectors (e.g., $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$):

$$\tilde{\mathbf{E}}(z, t) = \tilde{E}_0 e^{i(zt-\omega t)} \sum_i \Big(\hat{\mathbf{x}}\cos\theta_i + \hat{\mathbf{y}}\sin\theta_i\Big), \tag{52}$$

when the polarizer is parallel to $\hat{\mathbf{y}}$, the y components vanish. Hence, the left total electric waves are:

$$\tilde{\mathbf{E}}_{\text{fil}}(z, t) = \sum_i \hat{\mathbf{x}}\tilde{E}_0 e^{i(zt-\omega t)} \cos\theta_i. \tag{53}$$

If the polarizer's transmission axis is aligned with $\hat{\mathbf{x}}$, the y-components are filtered out. The resulting total electric field is given by:

3.3 Energy and Momentum

We consider plane wave in terms of $\mathbb{R}$, i.e.

$$\begin{cases} \mathbf{E} = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \phi) \\ \mathbf{B} = \dfrac{1}{c}\hat{\mathbf{k}} \times \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \phi) \end{cases} \Rightarrow B = \sqrt{\mu_0 \epsilon_0} E. \tag{54}$$

There are three quantities that we can discuss:

1. **Average Energy Density**: The energy density is given by

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) = \frac{1}{2}\left[\epsilon_0 E^2 + \frac{1}{\mu_0}\left(\mu_0\epsilon_0 E^2\right)\right] = \epsilon_0 E^2. \tag{55}$$

By (54), the square of amplitude of the electric field is

$$E^2 = E_0^2\cos^2(\mathbf{k}\cdot\mathbf{r} - \omega t + \phi). \tag{56}$$

Then the average of (55) over a complete cycle:

$$\langle u\rangle = \frac{1}{T}\epsilon_0 E_0^2\int_0^T \cos^2(\cdots)\,\mathrm{d}t = \frac{1}{2}\epsilon_0 E_0^2. \tag{57}$$

2. **Average Energy Flux and Intensity**: The Poynting vector is

$$\mathbf{S} = \frac{1}{\mu_0}\mathbf{E}\times\mathbf{B},$$

and we the Poynting vector propagates along a specific direction, which is same as the that of propagation of waves:

$$\begin{aligned}
\mathbf{S} &= \frac{1}{\mu_0}\mathbf{E}\times\mathbf{B} = \frac{1}{\mu_0 c}\mathbf{E}_0\times\left(\hat{\mathbf{k}}\times\mathbf{E}_0\right)\cos^2(\cdots)\\
&= \frac{1}{\mu_0 c}\hat{\mathbf{k}}E_0^2\cos^2(\cdots) = \frac{\epsilon_0}{\mu_0\epsilon_0 c}\hat{\mathbf{k}}E_0^2\cos^2(\cdots)\\
&= c\epsilon_0\hat{\mathbf{k}}E_0^2\cos^2(\mathbf{k}\cdot\mathbf{r} - \omega t + \phi).
\end{aligned}$$

Therefore, the average is

$$\langle\mathbf{S}\rangle = \frac{1}{2}c\hat{\mathbf{k}}\epsilon_0 E_0^2 = c\hat{\mathbf{k}}\left(\frac{1}{2}\epsilon_0 E_0^2\right) = \mathbf{c}\langle u\rangle. \tag{58}$$

We define the average power per unit area transported by an electromagnetic wave is **intensity**(I):

$$I \equiv \langle S\rangle = c\left(\frac{1}{2}\epsilon_0 E_0^2\right) = c\langle u\rangle. \tag{59}$$

3. **Average Momentum Density**: The momentum per unit volume $\mathbf{g}$ is given by:

$$\mathbf{g} = \mu_0\epsilon_0\mathbf{S} = \frac{1}{c^2}\mathbf{S}, \tag{60}$$

So, by (58), the average of $\mathbf{g}$ is:

$$\langle\mathbf{g}\rangle = \frac{1}{c^2}\langle\mathbf{S}\rangle = \hat{\mathbf{k}}\frac{1}{c}\left(\frac{1}{2}\epsilon_0 E_0^2\right). \tag{61}$$

4 Reflection and Transmission in Different Matters

Consider a linear media, and recall that the permittivity(ϵ) and the permeability(μ) are

$$\epsilon = (1 + \chi_e)\epsilon_0, \quad \mu = (1 + \chi_m)\mu_0,$$

hence the displacement field() and auxiliary field are

$$\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{B} = \mu\mathbf{H}.$$

So, when the media is in a source-free($\rho_f = |\mathbf{J}_f| = 0$) space, Maxwell's equations of matters become:

$$\begin{cases} \boldsymbol{\nabla} \cdot \mathbf{D} = 0 \\ \boldsymbol{\nabla} \cdot \mathbf{B} = 0 \\ \boldsymbol{\nabla} \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \boldsymbol{\nabla} \times \mathbf{H} = \dfrac{\partial \mathbf{D}}{\partial t} \end{cases} \quad \Rightarrow \quad \begin{cases} \boldsymbol{\nabla} \cdot \mathbf{E} = 0 \\ \boldsymbol{\nabla} \cdot \mathbf{B} = 0 \\ \boldsymbol{\nabla} \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \boldsymbol{\nabla} \times \mathbf{B} = \mu\epsilon\dfrac{\partial \mathbf{E}}{\partial t} \end{cases}, \tag{62}$$

This will happen only when μ and ϵ are constant, which means that the propagation is in the same media, then Maxwell's equations will be our familiar forms. So, the wave equation becomes

$$\nabla^2\mathbf{E} = \frac{1}{v^2}\frac{\partial^2\mathbf{E}}{\partial t^2}, \quad \nabla^2\mathbf{B} = \frac{1}{v^2}\frac{\partial^2\mathbf{B}}{\partial t^2}, \quad v = \frac{1}{\sqrt{\mu\epsilon}}. \tag{63}$$

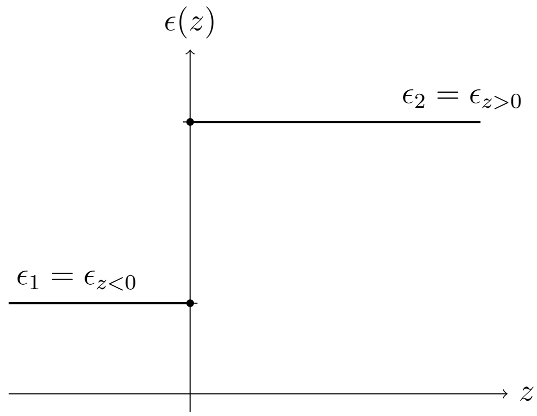
Comparing the speed of light and the speed of the electromagnetic wave in a linear medium, we have

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} > v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{(1+\chi_e)(1+\chi_m)}\sqrt{\mu_0\epsilon_0}} = \frac{c}{\sqrt{(1+\chi_e)(1+\chi_m)}}. \tag{64}$$

So, the speed in a linear medium is denoted by:

$$\boxed{v = \frac{c}{n}}, \quad n \equiv \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} = \sqrt{(1+\chi_e)(1+\chi_m)}. \tag{65}$$

By (64), the speed in a medium is lower than that in a vacuum since the medium may slow down the speed of propagation due to some friction or polarization lag. What if the permittivity and the permeability are not constant? For example.



Then we have

$$0 = \boldsymbol{\nabla} \cdot \mathbf{D} = \boldsymbol{\nabla} \cdot \Big(\epsilon(\mathbf{r})\mathbf{E}(\mathbf{r}, t)\Big) = \epsilon(\mathbf{r})\boldsymbol{\nabla} \cdot \mathbf{E} + \mathbf{E} \cdot \underbrace{\Big(\boldsymbol{\nabla}\epsilon(\mathbf{r})\Big)}_{\text{Not zero}},$$

and in this case, the gradient of permittivity function of z is not zero, the vertical axis represent the interface between two matter. The gradient is 0 when $z < 0$ or $z > 0$, however, there's a jump at $z = 0$. So, the delta function comes out:

$$\boldsymbol{\nabla}\epsilon(z) = \big(\hat{\mathbf{x}}\partial_x + \hat{\mathbf{y}}\partial_y + \hat{\mathbf{z}}\partial_z\big)\epsilon(z) = \hat{\mathbf{z}}\partial_z\epsilon(z) = \hat{\mathbf{z}}(\epsilon_2 - \epsilon_1)\delta(z). \tag{66}$$

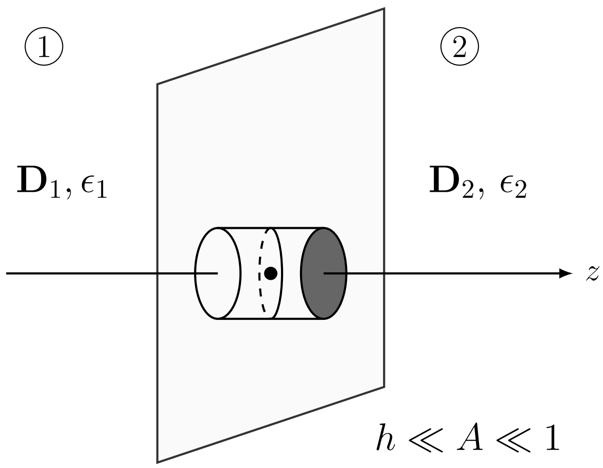
The integral shows the distribution of the permittivity:

$$\underbrace{\int_{0-\varepsilon}^{0+\varepsilon} \mathrm{d}z \, \big(\partial_z\epsilon(z)\big)}_{=\epsilon(0+\varepsilon)-\epsilon(0-\epsilon)} = (\epsilon_2 - \epsilon_1) \int_{0-\varepsilon}^{0+\varepsilon} \mathrm{d}z \, \delta(z) = \epsilon_2 - \epsilon_1, \tag{67}$$

where $\varepsilon > 0$ and $\epsilon(0 + \varepsilon) = \epsilon_2$, $\epsilon(0 - \varepsilon) = \epsilon_1$.

4.1 Boundary Condition

Consider a pillbox on the interface, with the area A and the height h, as shown below:



1. $D_1^\perp = D_2^\perp$:
   By Gauss's law and divergence theorem, we have:
   $$\int \mathrm{d}\tau\, \boldsymbol{\nabla} \cdot \mathbf{D} = 0 = \oint \mathrm{d}\mathbf{a} \cdot \mathbf{D}, \tag{68}$$
   since the height h is infinitesimal, so the area on the side of pillbox is 0, (68) can be written as
   $$\big(\mathbf{D}_1 \cdot \hat{\mathbf{n}}_1\big)A + \big(\mathbf{D}_2 \cdot \hat{\mathbf{n}}_2\big)A = 0.$$
   Note that the area vectors are outward, so this yields:
   $$-D_1^\perp + D_2^\perp = 0 \Rightarrow D_1^\perp = D_2^\perp \tag{69}$$

2. $B_1^\perp = B_2^\perp$:
   As the same reason of 1., we have:
   $$0 = \int \mathrm{d}\tau\, \boldsymbol{\nabla} \cdot \mathbf{B} = \oint \mathrm{d}\mathbf{a} \cdot \mathbf{B} = \big(\mathbf{B}_1 \cdot \hat{\mathbf{n}}_1\big)A + \big(\mathbf{B}_2 \cdot \hat{\mathbf{n}}_2\big)A \Rightarrow B_1^\perp = B_2^\perp.$$

3. $\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel$:
   Consider a Ampèrian loop with thin width(w), Stokes' Theorem gives
   $$\int \mathrm{d}\mathbf{a} \cdot \boldsymbol{\nabla} \times \mathbf{E} = \oint \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{E} = -\frac{d}{dt}\int \mathrm{d}\mathbf{a} \cdot \mathbf{B} \to 0 \tag{70}$$
   since $lw \to 0$ when the width $w \to 0$, which means that the magnetic flux vanishes. And the closed integral indicates the opposite directions of each side, therefore
   $$\oint \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{E} = \mathbf{E}_1 \cdot \boldsymbol{\ell} - \mathbf{E}_2 \cdot \boldsymbol{\ell} = 0 \Rightarrow \mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel. \tag{71}$$

4. $\mathbf{H}_1^{\parallel} = \mathbf{H}_2^{\parallel}$:

$$\int \mathrm{d}\mathbf{a} \cdot \boldsymbol{\nabla} \times \mathbf{H} = \oint \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{H} = \int \mathrm{d}\mathbf{a} \cdot \frac{\partial \mathbf{D}}{\partial t} \to 0,$$

hence,

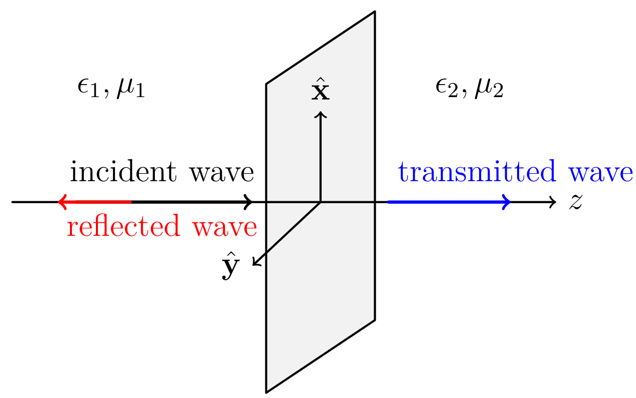
$$\oint \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{H} = \mathbf{H}_1 \cdot \boldsymbol{\ell} - \mathbf{H}_2 \cdot \boldsymbol{\ell} = 0 \Rightarrow \mathbf{H}_1^{\parallel} = \mathbf{H}_2^{\parallel}. \tag{72}$$

So, the four boundary conditions are:

$$\boxed{\begin{cases} D_1^{\perp} = D_2^{\perp},\ \epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp} \\ B_1^{\perp} = B_2^{\perp} \end{cases} \qquad \begin{cases} \mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel} \\ \mathbf{H}_1^{\parallel} = \mathbf{H}_2^{\parallel},\ \dfrac{1}{\mu_1}\mathbf{B}_1^{\parallel} = \dfrac{1}{\mu_2}\mathbf{B}_2^{\parallel} \end{cases}} \tag{73}$$

4.2 Normal Situation

Given an incident wave $\tilde{\mathbf{E}}_I$ and $\tilde{\mathbf{B}}_I$ which are perpendicular to the surface. When the wave contact the interface, there will be a reflected wave and a transmitted wave. As shown below



The incident wave is given by:

$$\begin{cases} \tilde{\mathbf{E}}_I = \tilde{\mathbf{E}}_{0_I} e^{i\left(\mathbf{k}_I \cdot \mathbf{r} - \omega_I t\right)} = \hat{\mathbf{x}} \tilde{E}_{0_I} e^{i\left(k_I z - \omega_I t\right)}, \\ \tilde{\mathbf{B}}_I = \dfrac{1}{v_1} \tilde{\mathbf{E}}_{0_I} e^{i\left(\mathbf{k}_I \cdot \mathbf{r} - \omega_I t\right)} = \hat{\mathbf{y}} \dfrac{1}{v_1} \tilde{E}_{0_I} e^{i\left(k_I z - \omega_I t\right)}. \end{cases} \qquad v_1 = \frac{1}{\sqrt{\mu_1 \epsilon_1}}, \tag{74}$$

where v_1 is the speed on the left region with μ_1 and ϵ_1. The orientations $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ here are chosen arbitrarily; as long as the two directions are mutually orthogonal, the x and y axes can be defined at will. This is because, with respect to the direction of propagation $\hat{\mathbf{z}}$, the wave possesses axial symmetry regardless of the coordinate system selected.
Then we can first determine the reflected wave and transmitted wave by their direction and the constants:

1. Reflected Wave: The direction of propagation of the reflected wave is opposite the incident wave, therefore:

$$\begin{cases} \tilde{\mathbf{E}}_R = \hat{\mathbf{x}} \tilde{E}_{0_R} e^{i\left(\mathbf{k}_R \cdot \mathbf{r} - \omega_R t\right)} = \hat{\mathbf{x}} \tilde{E}_{0_R} e^{i\left(-k_R z - \omega_R t\right)}, \\ \tilde{\mathbf{B}}_R = -\hat{\mathbf{y}} \dfrac{1}{v_1} \tilde{E}_{0_R} e^{i\left(\mathbf{k}_R \cdot \mathbf{r} - \omega_R t\right)} = -\hat{\mathbf{y}} \dfrac{1}{v_1} \tilde{E}_{0_R} e^{i\left(-k_R z - \omega_R t\right)}. \end{cases} \qquad v_1 = \frac{1}{\sqrt{\mu_1 \epsilon_1}}, \tag{75}$$

2. Transmitted Wave:The direction of propagation of the reflected wave is same as the incident wave, i.e., along $\hat{\mathbf{z}}$, therefore: beginequation

$$\begin{cases} \tilde{\mathbf{E}}_T = \hat{\mathbf{x}} \tilde{E}_{0_T} e^{i\left(\mathbf{k}_T \cdot \mathbf{r} - \omega_T t\right)} = \hat{\mathbf{x}} \tilde{E}_{0_T} e^{i\left(k_T z - \omega_T t\right)}, \\ \tilde{\mathbf{B}}_T = \hat{\mathbf{y}} \dfrac{1}{v_2} \tilde{E}_{0_T} e^{i\left(\mathbf{k}_T \cdot \mathbf{r} - \omega_T t\right)} = \hat{\mathbf{y}} \dfrac{1}{v_2} \tilde{E}_{0_T} e^{i\left(k_T z - \omega_T t\right)}. \end{cases} \qquad v_2 = \frac{1}{\sqrt{\mu_2 \epsilon_2}}. \tag{76}$$

To determine the relation between k_I, k_R, k_T and $\omega_I, \omega_R, \omega_T$, we need to consider the boundary condition $\mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel}$ on the interface. These waves must satisfy the boundary conditions at any moment, or, always satisfy. So:

$$\mathbf{E}_1^{\parallel} = \hat{\mathbf{x}} \tilde{E}_{0_I} e^{\overbrace{i(0x - \omega_I t)}^{\text{No } \hat{\mathbf{x}} \text{ propagation}}} + \hat{\mathbf{x}} \tilde{E}_{0_R} e^{-i\omega_R t} = \mathbf{E}_2^{\parallel} = \hat{\mathbf{x}} \tilde{E}_{0_T} e^{-i\omega_T t}. \tag{77}$$

This forces that

$$\omega_I = \omega_R = \omega_T \equiv \omega, \tag{78}$$

or we cannot eliminate the exponential term depending on time t to ensure that they always satisfy the condition. So, the frequency **Remain the Same Value**. Since the frequency $\omega = vk$, then $v_1 k_I = v_1 k_R = v_2 k_T$, that is

$$k_I = k_R \equiv k_1, \quad k_2 \equiv k_T = \frac{v_1}{v_2} k_1 = \frac{n_2}{n_1} k_1 = \sqrt{\frac{\mu_2 \epsilon_2}{\mu_1 \epsilon_1}} k_1. \tag{79}$$

After determining the frequency and the wave numbers, (77) gives that

$$\hat{\mathbf{x}} \tilde{E}_{0_I} e^{-i\omega t} + \hat{\mathbf{x}} \tilde{E}_{0_R} e^{-i\omega t} = \hat{\mathbf{x}} \tilde{E}_{0_T} e^{-i\omega t} \Rightarrow \tilde{E}_{0_I} + \tilde{E}_{0_R} = \tilde{E}_{0_T}, \tag{80}$$

furthermore, by the boundary condition $\mathbf{H}_1^{\parallel} = \mathbf{H}_2^{\parallel}$, then

$$\mathbf{H}_1^{\parallel} = \hat{\mathbf{y}} \frac{1}{\mu_1 v_1} \tilde{E}_{0_I} e^{-i\omega t} - \hat{\mathbf{y}} \frac{1}{\mu_1 v_1} \tilde{E}_{0_R} e^{-i\omega t} = \mathbf{H}_2^{\parallel} = \hat{\mathbf{y}} \frac{1}{\mu_2 v_2} \tilde{E}_{0_T} e^{-i\omega t},$$

and this yields

$$\tilde{E}_{0_I} - \tilde{E}_{0_R} = \frac{\mu_1 v_1}{\mu_2 v_2} \tilde{E}_{0_T}. \tag{81}$$

The sum of (80) and (81) is

$$2\tilde{E}_{0_I} = \left(1 + \frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}\right) \tilde{E}_{0_T},$$

therefore

$$\tilde{E}_{0_T} = \frac{2}{\left(1 + \dfrac{\mu_1 v_1}{\mu_2 v_2}\right)} \tilde{E}_{0_I}. \tag{82}$$

Substituting $\tilde{E}_{0_T}$ into (80), we can obtain the final relation between three waves:

$$\tilde{E}_{0_R} = \tilde{E}_{0_T} - \tilde{E}_{0_I} = \left(\frac{1 - \dfrac{\mu_1 v_1}{\mu_2 v_2}}{1 + \dfrac{\mu_1 v_1}{\mu_2 v_2}}\right) \tilde{E}_{0_I}. \tag{83}$$

We define a new constant $\beta = \mu_1 v_1 / \mu_2 v_2$. Note that $v = c/n$ in (65), so β can also be denoted by:

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1}. \tag{84}$$

Substituting β into (82) and (83), we can obtain:

$$\boxed{\begin{cases} \tilde{E}_{0_T} = \dfrac{2}{1+\beta} \tilde{E}_{0_I} \\ \tilde{E}_{0_R} = \dfrac{1-\beta}{1+\beta} \tilde{E}_{0_I}. \end{cases}} \tag{85}$$

There's another important property of the waves. By (59) and (85), the intensities of them are:

$$I_I = \frac{1}{2} \epsilon_1 v_1 \left|\tilde{E}_{0_I}\right|^2, \quad I_R = \frac{1}{2} \epsilon_1 v_1 \left(\frac{1-\beta}{1+\beta}\right)^2 \left|\tilde{E}_{0_I}\right|^2, \quad I_T = \frac{1}{2} \epsilon_2 v_2 \left(\frac{2}{1+\beta}\right)^2 \left|\tilde{E}_{0_I}\right|^2. \tag{86}$$

Note that we can rewrite the constant $\beta(\mu, v)$ in terms of $\beta(\epsilon, v)$. By (84), we obtain

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 \dfrac{1}{\sqrt{\mu_1 \epsilon_1}}}{\mu_2 \dfrac{1}{\sqrt{\mu_2 \epsilon_2}}} = \frac{\sqrt{\dfrac{\mu_1}{\epsilon_1}}}{\sqrt{\dfrac{\mu_2}{\epsilon_2}}} = \frac{1/\sqrt{\dfrac{\epsilon_1}{\mu_1}}}{1/\sqrt{\dfrac{\epsilon_2}{\mu_2}}} = \frac{\sqrt{\dfrac{\epsilon_2}{\mu_2}}}{\sqrt{\dfrac{\epsilon_1}{\mu_1}}},$$

which leads to:

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}. \tag{87}$$

Define the ratio of reflected intensity to the incident intensity as R, and the ratio of transmitted intensity to the incident intensity as T, which is

$$R \equiv \frac{I_R}{I_I}, \qquad T \equiv \frac{I_T}{I_I}. \tag{88}$$

Substituting (86) into it, then

$$R = \frac{\frac{1}{2}\epsilon_1 v_1 \left(\frac{1-\beta}{1+\beta}\right)^2 \left|\tilde{E}_{0_I}\right|^2}{\frac{1}{2}\epsilon_1 v_1 \left|\tilde{E}_{0_I}\right|^2} = \frac{\beta^2 - 2\beta + 1}{(\beta+1)^2}$$

$$T = \frac{\frac{1}{2}\epsilon_2 v_2 \left(\frac{2}{1+\beta}\right)^2 \left|\tilde{E}_{0_I}\right|^2}{\frac{1}{2}\epsilon_1 v_1 \left|\tilde{E}_{0_I}\right|^2} = \left(\frac{\epsilon_2 v_2}{\epsilon_1 v_1}\right)\frac{4}{(\beta+1)^2} = \frac{4\beta}{(\beta+1)^2}$$

hence,

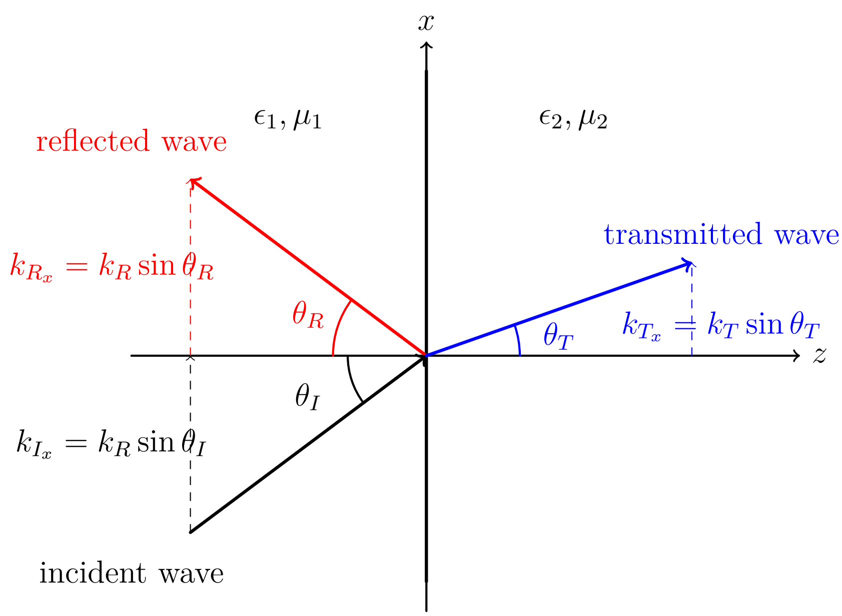
$$R + T = \frac{I_R}{I_I} + \frac{I_T}{I_I} = \frac{\beta^2 - 2\beta + 1}{(\beta+1)^2} + \frac{4\beta}{(\beta+1)^2} = \frac{(\beta+1)^2}{(\beta+1)^2} = 1, \tag{89}$$

we can also write

$$I_R + I_T = I_I. \tag{90}$$

4.3 Oblique Situation

An oblique monochromatic plane wave $\tilde{\mathbf{E}}_I, \tilde{\mathbf{B}}_I$ approaches from the left, giving rise to a reflected wave $\tilde{\mathbf{E}}_R, \tilde{\mathbf{B}}_R$ and a transmitted wave $\tilde{\mathbf{E}}_T, \tilde{\mathbf{B}}_T$. As shown below:



Where the propagation of the incident wave can be denoted by:

$$\mathbf{k}_I = \frac{\omega}{v_1}\Big(\hat{\mathbf{z}}\cos\theta_I + \hat{\mathbf{x}}\sin\theta_I\Big) \tag{91}$$

›

4.3.1 Determining the Angle and the Wave Number(Vector) By Geometry

Suppose that the waves are:

$$\text{Incident}: \tilde{\mathbf{E}}_I = \tilde{\mathbf{E}}_{0_I} e^{i(\mathbf{k}_I\cdot\mathbf{r}-\omega t)}, \quad \tilde{\mathbf{B}}_I = \frac{1}{v_1}(\hat{\mathbf{k}}_I \times \tilde{\mathbf{E}}_I).$$

$$\text{Reflected}: \tilde{\mathbf{E}}_R = \tilde{\mathbf{E}}_{0_R} e^{i(\mathbf{k}_R\cdot\mathbf{r}-\omega t)}, \quad \tilde{\mathbf{B}}_R = \frac{1}{v_1}(\hat{\mathbf{k}}_R \times \tilde{\mathbf{E}}_R).$$

$$\text{Transmitted}: \tilde{\mathbf{E}}_T = \tilde{\mathbf{E}}_{0_T} e^{i(\mathbf{k}_T\cdot\mathbf{r}-\omega t)}, \quad \tilde{\mathbf{B}}_T = \frac{1}{v_2}(\hat{\mathbf{k}}_T \times \tilde{\mathbf{E}}_T).$$

By (78), the frequency of waves are the same, namely

$$\omega = k_I v_1 = k_R v_1 = k_T v_2,$$

so the three wave numbers are related:

$$k_I = k_R = \frac{v_2}{v_1}k_T = \frac{n_1}{n_2}k_T. \quad \text{(Amplitude)} \tag{92}$$

In addition, the tangential components of the wave vectors are equal because the boundary conditions must hold at every point on the interface. This requires the phases of all waves to be identical along the boundary, which forces the tangential components of the wave vectors to be the same. When $z = 0$(boundary), we have

$$\mathbf{k}_I^{\parallel}\cdot\mathbf{r} = \mathbf{k}_R^{\parallel}\cdot\mathbf{r} = \mathbf{k}_T^{\parallel}\cdot\mathbf{r}, \ (z = 0, \text{at any time}) \tag{93}$$

which can be written as

$$k_{I_x}x + k_{I_y}y = k_{R_x}x + k_{R_y}y = k_{T_x}x + k_{T_y}y \tag{94}$$

for all x, y. Then with different conditions:

$$\begin{cases} k_{I_y} = k_{R_y} = k_{T_y} \quad (x = 0, z = 0) \\ \\ k_{I_x} = k_{R_x} = k_{T_x} \quad (y = 0, z = 0). \end{cases} \tag{95}$$

By the second equation, the x component of each wave number is

$$k_{I_x} = k_I\sin\theta_I, \quad k_{R_x} = k_R\sin\theta_R, \quad k_{T_x} = k_T\sin\theta_T. \tag{96}$$

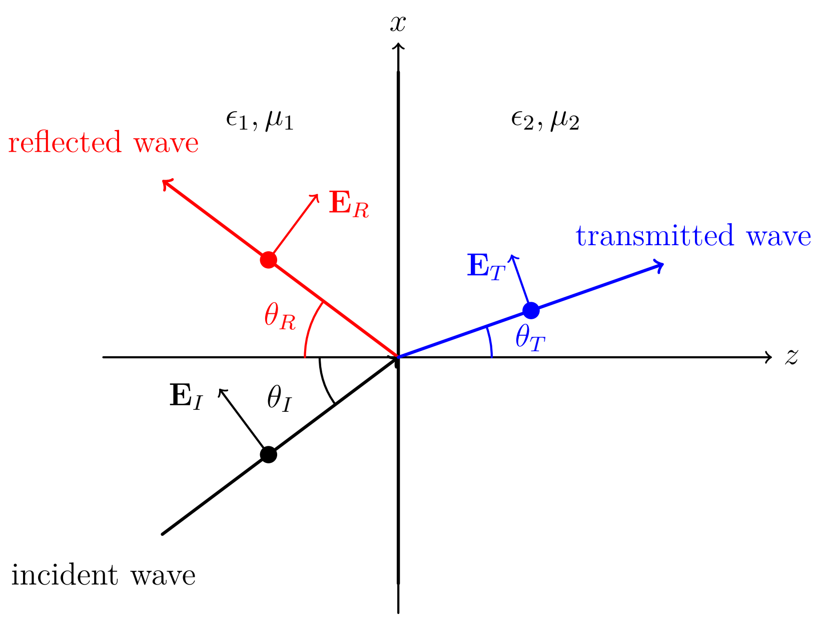
So, there are two crucial rule that the EM waves have:

1. **<u>Law of Reflection(1st Law)</u>**: Since $k_I = k_R$, the equations (96) can be written as
$$k_I \sin\theta_I = k_R \sin\theta_R \Rightarrow \sin\theta_I = \sin\theta_R.$$
This yields:
$$\theta_I = \theta_R. \tag{97}$$

2. **<u>Snell's Law(2nd Law)</u>**:By substituting (92) into (96), it becomes
$$k_I \sin\theta_I = k_T \sin\theta_T \Rightarrow k_I \sin\theta_I = \frac{n_2}{n_1} k_I \sin\theta_T,$$
then we can obtain
$$\frac{n_1}{n_2} = \frac{\sin\theta_T}{\sin\theta_I} \Leftrightarrow n_1 \sin\theta_I = n_2 \sin\theta_2. \tag{98}$$

4.3.2 Determining the Amplitude of the Field By Polarization and B.C.

Consider a direction of polarization of the incident wave which is parallel to xz plane. In fact, **we can choose arbitrary direction of polarization**, but to simplify the problem, we usually pick this one, as shown below:



We can choose some reference point on the wave, here, to simplify the equation, we usually choose $p-$plane, which is parallel to xz plane. The boundary conditions in this case become:

$$\begin{cases} \text{(i) } \epsilon_1(\tilde{\mathbf{E}}_{0_I} + \tilde{\mathbf{E}}_{0_R})_z = \epsilon_2(\tilde{\mathbf{E}}_{0_T})_z \\ \text{(ii) } (\tilde{\mathbf{B}}_{0_I} + \tilde{\mathbf{B}}_{0_R})_z = (\tilde{\mathbf{B}}_{0_T})_z \\ \text{(iii) } (\tilde{\mathbf{E}}_{0_I} + \tilde{\mathbf{E}}_{0_R})_{x,y} = (\tilde{\mathbf{E}}_{0_T})_{x,y} \\ \text{(iv) } \dfrac{1}{\mu_1}(\tilde{\mathbf{B}}_{0_I} + \tilde{\mathbf{B}}_{0_R})_{x,y} = \dfrac{1}{\mu_2}(\tilde{\mathbf{B}}_{0_T})_{x,y}, \end{cases} \tag{99}$$

by condition (i), we have

$$\epsilon_1(-\tilde{E}_{0_I}\sin\theta_I + \tilde{E}_{0_R}\sin\theta_R) = \epsilon_2(-\tilde{E}_{o_T}\sin\theta_T). \tag{100}$$

There's no magnetic field along $\hat{\mathbf{z}}$, then (ii) dropped. Furthermore, (iii) gives that

$$\tilde{E}_{0_I}\cos\theta_I + \tilde{E}_{0_R}\cos\theta_R = \tilde{E}_{0_T}\cos\theta_T \Rightarrow \tilde{E}_{0_I} + \tilde{E}_{0_R} = \alpha\tilde{E}_{0_T}, \tag{101}$$

where

$$\alpha \equiv \frac{\cos\theta_T}{\cos\theta_I}. \tag{102}$$

And the last one, we know that $\tilde{\mathbf{B}}_R = -\hat{\mathbf{k}} \times \tilde{\mathbf{E}}$, so (iv) leads to

$$\frac{1}{\mu_1 v_1}(\tilde{E}_{0_I} - \tilde{E}_{0_R}) = \frac{1}{\mu_2 v_2}\tilde{E}_{0_T} \Rightarrow \tilde{E}_{0_I} - \tilde{E}_{0_R} = \frac{\mu_1 v_1}{\mu_2 v_2}\tilde{E}_{0_T} = \beta\tilde{E}_{0_T}. \tag{103}$$

By (101) and (103), the summation gives that

$$\tilde{E}_{0_T} = \left(\frac{2}{\alpha+\beta}\right)\tilde{E}_{0_I}. \tag{104}$$

Substituting $\tilde{E}_{0_T}$ into (101), we can obtain:

$$\tilde{E}_{0_R} = \left(\frac{2\alpha}{\alpha+\beta}\right)\tilde{E}_{0_I} - \tilde{E}_{0_I} = \left(\frac{\alpha-\beta}{\alpha+\beta}\right)\tilde{E}_{0_I}. \tag{105}$$

After these analysis, we can compare the above configurations:

$$\boxed{\text{Normal}: \begin{cases} \tilde{E}_{0_T} = \dfrac{2}{1+\beta}\tilde{E}_{0_I} \\ \tilde{E}_{0_R} = \dfrac{1-\beta}{1+\beta}\tilde{E}_{0_I} \end{cases} \qquad \text{Oblique}: \begin{cases} \tilde{E}_{0_T} = \dfrac{2}{\alpha+\beta}\tilde{E}_{0_I} \\ \tilde{E}_{0_R} = \dfrac{\alpha-\beta}{\alpha+\beta}\tilde{E}_{0_I} \end{cases}}. \tag{106}$$

with

$$\boxed{\begin{cases} \alpha = \dfrac{\cos\theta_T}{\cos\theta_I} \\ \beta = \dfrac{\mu_1 v_1}{\mu_2 v_2} = \dfrac{\mu_1 n_2}{\mu_2 n_1}. \end{cases}} \tag{107}$$

So, the SOP of solving the EM waves problem under oblique configuration is:

- **Step 1**: Set up the geometry (define the interface and coordinate system). Choose the plane of incidence and identify the normal direction.
- **Step 2**: Determine all wave vectors using phase matching:

  $$\mathbf{k}_I^{\parallel} = \mathbf{k}_R^{\parallel} = \mathbf{k}_T^{\parallel}$$

  This gives the propagation directions and angles (reflection law and Snell's law).
- **Step 3**: Choose the polarization (s or p polarization). This selects a basis and simplifies the vector structure of the fields.
- **Step 4**: Determine the direction of the fields using the transverse condition:

  $$\mathbf{k} \cdot \mathbf{E} = 0$$

  Then obtain the magnetic field from

  $$\mathbf{B} = \frac{1}{\omega}\mathbf{k} \times \mathbf{E}$$
- **Step 5**: Write the full plane wave solutions with unknown amplitudes:

  $$\mathbf{E}_I,\ \mathbf{E}_R,\ \mathbf{E}_T$$

  The only unknowns are the scalar amplitudes.
- **Step 6**: Apply boundary conditions at the interface:

  $$\mathbf{E}_I^{\parallel} + \mathbf{E}_R^{\parallel} = \mathbf{E}_T^{\parallel}$$

  $$\mathbf{H}_I^{\parallel} + \mathbf{H}_R^{\parallel} = \mathbf{H}_T^{\parallel}$$
- **Step 7**: Solve the resulting linear system to obtain the reflection and transmission coefficients (amplitudes).

5 Electromagnetic Waves in Conductors

5.1 Decay EM Waves in a Conductor

When an EM waves propagates in a conductor, there must be charges distribution. In addition, the electric field induces current to generate a current density $\mathbf{J}_f$, which satisfies

$$\mathbf{J}_f = \sigma\mathbf{E}, \tag{108}$$

where σ is **conductivity**. Assume that the wave is in a linear medium, then Maxwell's equations give:

$$\begin{cases} \boldsymbol{\nabla} \cdot \mathbf{E} = \dfrac{\rho_f}{\epsilon} \\ \boldsymbol{\nabla} \cdot \mathbf{B} = 0 \\ \boldsymbol{\nabla} \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\ \boldsymbol{\nabla} \times \mathbf{B} = \underbrace{\mu\sigma\mathbf{E}}_{\mu\mathbf{J}_f} + \mu\epsilon\dfrac{\partial \mathbf{E}}{\partial t}. \end{cases} \tag{109}$$

What will happen when a wave propagates in a conductor? The continuity equation gives:

$$\frac{\partial \rho_f}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J}_f = \frac{\partial \rho_f}{\partial t} + \boldsymbol{\nabla} \cdot (\sigma \mathbf{E}) = 0,$$

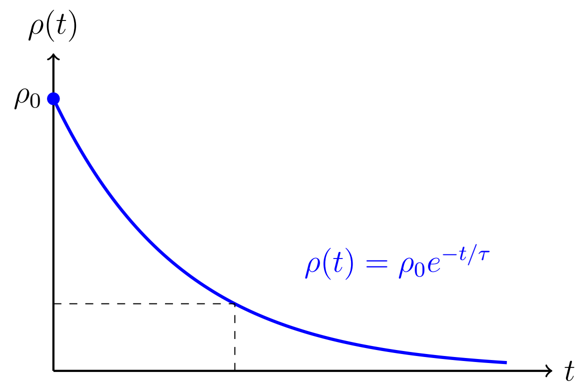
and by Gauss's law, this becomes a differential equation:

$$\frac{\partial \rho_f}{\partial t} + \boldsymbol{\nabla} \cdot (\sigma \mathbf{E}) = \frac{d}{dt}\rho_f(\mathbf{r}, t) + \frac{\sigma}{\epsilon}\rho_f(\mathbf{r}, t) = 0. \tag{110}$$

The solution is

$$\rho_f(\mathbf{r}, t) = \rho_0 e^{-(\sigma/\epsilon)t} = \rho_0 e^{-t/\tau}, \tag{111}$$

where $\tau \equiv \epsilon/\sigma$. This tells us that the free charges will exponentially decay in a conductor. When the conductivity is large, it means that the medium conducts strongly, making the charges ρ_f "escape". As shown below:



Physical Meaning of Charge Density Decay in a Conductor

In a conducting medium, the free charge density satisfies

$$\rho_f(t) = \rho_0 e^{-t/\tau}, \qquad \tau = \frac{\epsilon}{\sigma}.$$

This result implies that any excess free charge in the conductor will decay exponentially in time. The physical mechanism behind this behavior can be understood as follows:

- A nonzero charge density ρ_f generates an electric field $\mathbf{E}$ via Gauss's law.
- The electric field induces a current $\mathbf{J} = \sigma \mathbf{E}$ according to Ohm's law.
- This current transports charge away, reducing the original charge density.

Therefore, the system naturally evolves toward a state with no accumulated free charge:

$$\rho_f \to 0.$$

Implication for the Electric Field Since Gauss's law relates the divergence of the electric field to charge density,

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon},$$

the decay of ρ_f implies that

$$\nabla \cdot \mathbf{E} \to 0.$$

Physically, this means that any electric field generated by free charges cannot be sustained in the conductor. The medium continuously redistributes charge in such a way as to neutralize internal electric fields.

Comparison with Source-Free Fields in Vacuum In vacuum, the source-free condition is given by

$$\rho = 0, \qquad \mathbf{J} = 0.$$

This represents a fundamentally different situation:

- There are no charges present to generate fields.
- There are no currents to respond to the fields.
- The electromagnetic field exists independently and can propagate without being altered by the medium.

Key Physical Difference

- In vacuum, $\rho = 0$ because there are no charges at all.
- In a conductor, $\rho_f \to 0$ because charges actively move to cancel any imbalance.

Thus, in a conductor, the vanishing of charge density is not a passive condition but the result of an active dynamical process. The medium continuously reacts to electric fields by redistributing charge, preventing the buildup of sustained electric field configurations.

Summary

- Charge density decay indicates that conductors cannot support persistent charge accumulation.
- Any electric field associated with free charges is rapidly neutralized.
- Unlike vacuum, where fields propagate freely, a conductor actively responds to fields and suppresses charge-induced field structures.

The EM wave equation now is modified, since

$$\underbrace{\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{E})}_{= \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{E}) - \nabla^2 \mathbf{E}} = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \mathbf{B}) = -\frac{\partial}{\partial t}\left(\mu\sigma\mathbf{E} + \mu\epsilon\frac{\partial \mathbf{E}}{\partial t}\right). \tag{112}$$

Consider a steady state, namely $\boldsymbol{\nabla} \cdot \mathbf{E} \to 0$(after ρ_f decays to 0, very fast). So, the wave equation becomes

$$\nabla^2 \mathbf{E} = \mu\epsilon\frac{\partial^2 \mathbf{E}}{\partial t^2} + \underbrace{\sigma\mu\frac{\partial \mathbf{E}}{\partial t}}_{\text{Damp}} \Rightarrow \nabla^2 \mathbf{E} - \mu\epsilon\frac{\partial^2 \mathbf{E}}{\partial t^2} = \sigma\mu\frac{\partial \mathbf{E}}{\partial t}. \tag{113}$$

<u>Ansatz</u>: the solution may be

$$\tilde{\mathbf{E}} = \tilde{\mathbf{E}}(\tilde{\mathbf{k}})e^{i(\tilde{\mathbf{k}}\cdot\mathbf{r}-\omega t)} \quad \tilde{\mathbf{B}} = \frac{\tilde{k}}{\omega}\left[\hat{\tilde{\mathbf{k}}} \times \tilde{\mathbf{E}}(\tilde{\mathbf{k}})e^{i(\tilde{\mathbf{k}}\cdot\mathbf{r}-\omega t)}\right], \tag{114}$$

and substitute (114) into (113), we can obtain a algebraic equation:

$$-\tilde{k}^2 + \mu\epsilon\omega^2 = -i\mu\sigma\omega \Rightarrow \tilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega. \tag{115}$$

This time, the wave vector is complex now.[5] Let [6]

$$\tilde{\mathbf{k}} = \mathbf{k}_r + i\mathbf{k}_i, \qquad \tilde{k} = k_r + ik_i. \tag{116}$$

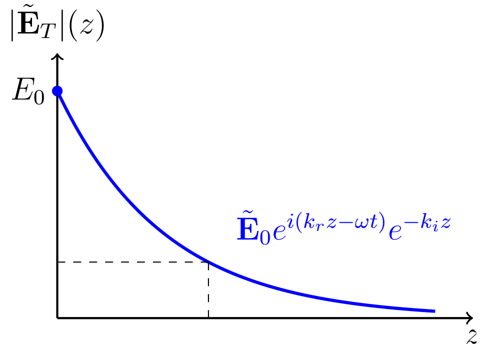
Assume the EM waves propagates along $\hat{\mathbf{z}}$, then the transmitted wave is:

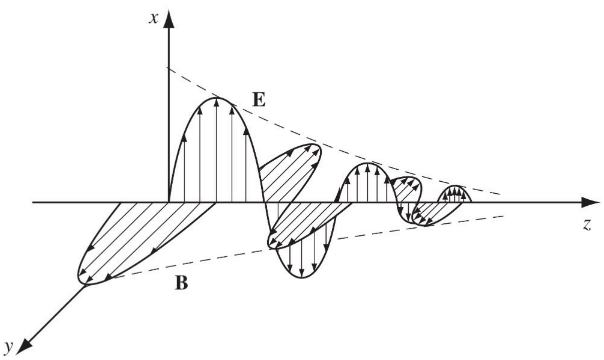
$$\tilde{\mathbf{E}}_T(\tilde{\mathbf{k}}) = \tilde{\mathbf{E}}_0 e^{i(\tilde{\mathbf{k}}\cdot\mathbf{r}-\omega t)} = \tilde{\mathbf{E}}_0 e^{i(\mathbf{k}_r+i\mathbf{k}_i)\cdot\mathbf{r}-i\omega t} = \tilde{\mathbf{E}}_0 e^{i(k_r z-\omega t)} e^{-k_i z}. \tag{117}$$

where k_i must be greater than 0, or the field will exponentially increase. (117) shows that the transmitted wave has oscillating part and the decay part. The oscillation phase of an electromagnetic wave evolves over time; however, its amplitude diminishes with increasing distance from the source. As shown below

[5] We cannot let the ω be complex because the solution have the term $e^{\omega_i t}$, which cannot approach the steady state.

[6] r, i represent real part and imaginary part, respectively.





By (117) and (114), the magnetic wave is therefore

$$\tilde{\mathbf{B}}_T = \frac{\tilde{k}}{\omega}\left[\hat{\tilde{\mathbf{k}}} \times \tilde{\mathbf{E}}_0 e^{i(k_r z - \omega t)}\right] e^{-k_i z}. \tag{118}$$

To evaluate k_r and k_i, we substitute (116) into (115), we have:

$$\tilde{k}^2 = k_r^2 + 2ik_r k_i - k_i^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega,$$

and compare the real part and the imaginary part, we obtain

$$\begin{cases} \text{(i) } k_r^2 - k_i^2 = \mu\epsilon\omega^2 \\ \\ \text{(ii) } 2k_i k_r = \mu\sigma\omega. \end{cases} \tag{119}$$

(ii) indicates that $k_i = \mu\sigma\omega/2k_r$, so substituting k_i into (i) can obtain

$$k_r^2 - \frac{\mu^2\sigma^2\omega^2}{4k_r^2} = \mu\epsilon\omega^2 \Rightarrow 4k_r^4 - 4\mu\epsilon\omega^2 k_r^2 - \mu^2\sigma^2\omega^2 = 0, \tag{120}$$

use the formula:

$$k_r^2 = \frac{4\mu\epsilon\omega^2 + \sqrt{16\mu^2\epsilon^2\omega^4 + 16\mu^2\sigma^2\omega^2}}{8} = \frac{4\mu\epsilon\omega^2 + 4\mu\epsilon\omega^2\sqrt{1 + (\cdots)}}{8}$$

$$= \frac{\mu\epsilon\omega^2}{2}\left[1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}\right],$$

therefore, the real part k_r is:

$$k_r = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} + 1\right]}. \tag{121}$$

Similarly,

$$k_i = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} - 1\right]}. \tag{122}$$

5.2 Boundary Conditions and Reflection

Since the subject of discussion is EM wave in a conductor now, so the surface charge and current exist. Hence, the boundary conditions become:

$$\begin{cases} \text{(i)}\ \epsilon_1 E_1 - \epsilon_2 E_2 = \sigma_f \\ \\ \text{(ii)}\ B_1^\perp = B_2^\perp \\ \\ \text{(iii)}\ \mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel \\ \\ \text{(iv)}\ \dfrac{1}{\mu_1}\mathbf{B}_1^\parallel - \dfrac{1}{\mu_2}\mathbf{B}_2^\parallel = \mathbf{K}_f \times \hat{\mathbf{n}}. \end{cases} \tag{123}$$

(iv) is the most important condition, recall that:

$$\oint_{\mathcal{P}} \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{H} = I_{\text{enc}} + \frac{d}{dt}\int_{\mathcal{S}} \mathrm{d}\mathbf{a} \cdot \mathbf{D} \Rightarrow \mathbf{H}_1 \cdot \boldsymbol{\ell} - \mathbf{H}_2 \cdot \boldsymbol{\ell} = I_{f_{\text{enc}}}, \tag{124}$$

where $I_{f_{\text{enc}}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \boldsymbol{\ell}) = (\mathbf{K}_f \times \hat{\mathbf{n}}) \cdot \boldsymbol{\ell}$, so (iv) holds. By these new boundary conditions, assume that there's an EM wave with electric field oscillates along $\hat{\mathbf{x}}$ and propagates to $\hat{\mathbf{z}}$, so we write down:

$$\begin{aligned} &\text{Incident}: \tilde{\mathbf{E}}_I(z,t) = \hat{\mathbf{x}}\tilde{E}_{0_I} e^{i(k_1 z - \omega t)}, \quad \tilde{\mathbf{B}}_I = \hat{\mathbf{y}}\frac{1}{v_1}\tilde{E}_{0_I} e^{i(k_1 z - \omega t)}. \\ \\ &\text{Reflected}: \tilde{\mathbf{E}}_R(z,t) = \hat{\mathbf{x}}\tilde{E}_{0_R} e^{i(-k_1 z - \omega t)}, \quad \tilde{\mathbf{B}}_R = -\hat{\mathbf{y}}\frac{1}{v_1}\tilde{E}_{0_R} e^{i(-k_1 z - \omega t)}. \\ \\ &\text{Transmitted}: \tilde{\mathbf{E}}_T(z,t) = \hat{\mathbf{x}}\tilde{E}_{0_T} e^{i(\tilde{k}_2 z - \omega t)}, \quad \tilde{\mathbf{B}}_T = \hat{\mathbf{y}}\frac{\tilde{k}_2}{\omega}\tilde{E}_{0_T} e^{i(\tilde{k}_2 z - \omega t)}. \end{aligned} \tag{125}$$

B.C. (iii) gives that

$$\tilde{E}_{0_I} + \tilde{E}_{0_R} = \tilde{E}_{0_T}, \tag{126}$$

and if there's no current on the interface, i.e. $\mathbf{K}_f = \mathbf{0}$, then (iv) gives:

$$\frac{1}{\mu_1 v_1}(\tilde{E}_{0_I} - \tilde{E}_{0_R}) = \frac{\tilde{k}_2}{\mu_2 \omega}\tilde{E}_{0_T} \Rightarrow \tilde{E}_{0_I} - \tilde{E}_{0_R} = \tilde{\beta}\tilde{E}_{0_T}, \tag{127}$$

where

$$\tilde{\beta} = \frac{\mu_1 v_1}{\mu_2 \omega}\tilde{k}_2. \tag{128}$$

The summation of (126) and (127) gives the two solutions:

$$\boxed{\begin{cases} \tilde{E}_{0_T} = \left(\dfrac{2}{1+\tilde{\beta}}\right)\tilde{E}_{0_I} \\ \tilde{E}_{0_R} = \left(\dfrac{1-\tilde{\beta}}{1+\tilde{\beta}}\right)\tilde{E}_{0_I}. \end{cases}} \tag{129}$$

Similar to (85)! But this $\tilde{\beta}$ is complex. If it's a perfect conductor, the conductivity σ is fucking large($\to\infty$) to make $\rho(z,t)$ decay infinitely fast. That is,

$$\tilde{k}_2 = \lim_{\sigma\to\infty}\omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}+1\right]} + i\omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}-1\right]} \to \infty,$$

this makes:

$$\lim_{\sigma\to\infty}\tilde{E}_{0_T} = \lim_{\tilde{\beta}\to\infty}\left(\frac{2}{1+\tilde{\beta}}\right)\tilde{E}_{0_I} = 0, \tag{130}$$

and

$$\lim_{\sigma\to\infty}\tilde{E}_{0_R} = \lim_{\tilde{\beta}\to\infty}\left(\frac{1-\tilde{\beta}}{1+\tilde{\beta}}\right)\tilde{E}_{0_I} = -\tilde{E}_{0_I}. \tag{131}$$

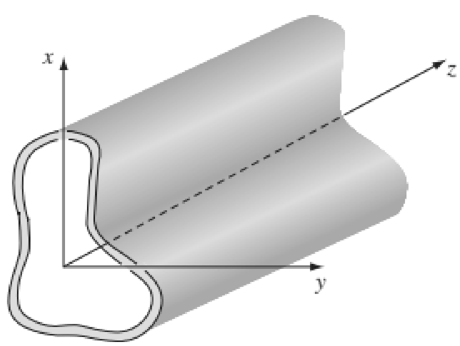
In conclusion, the wave hit the perfect conductor is totally reflected, with 180^o phase shift:

$$\tilde{E}_{0_T} = 0, \quad \tilde{E}_{0_R} = -\tilde{E}_{0_I}. \tag{132}$$

6 Guided Waves

6.1 General Solution to Guided Waves

Consider an electromagnetic wave confined in a **Wave Guide** shown below:



We regard the wave guide in this chapter as a perfect conductor where the electric field inside it is 0:

$$\mathbf{E}_{\text{Conductor}} = \mathbf{0},$$

with boundary condition (123)-(iii), it's obvious that $\mathbf{E}^{\parallel} = \mathbf{0}$. By Faraday's law, if the magnetic field starts out from 0, it will remain so since

$$\boldsymbol{\nabla} \times \mathbf{E}^{\parallel} = \boldsymbol{\nabla} \times \mathbf{0} = -\frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}.$$

Therefore, the wave in a wave guide must satisfy the boundary conditions:

$$\begin{cases} \mathbf{E}^{\parallel} = \mathbf{0} \\ \\ B^{\perp} = 0. \end{cases} \tag{133}$$

As the figure showed, we assume that the EM wave propagates along $\hat{\mathbf{z}}$[7]

Ansatz: The waves are

$$\begin{cases} \tilde{\mathbf{E}}(x,y,z,t) = \tilde{\mathbf{E}}_0(x,y)e^{i(kz-\omega t)}, \\ \\ \tilde{\mathbf{B}}(x,y,z,t) = \tilde{\mathbf{B}}_0(x,y)e^{i(kz-\omega)}. \end{cases} \tag{134}$$

However, the wave are not always transverse(just along $\hat{\mathbf{x}}$ or $\hat{\mathbf{y}}$) since it's strictly constrained in the wave guide made of perfect conductor. So, it may have z-component:

$$\tilde{\mathbf{E}}_0 = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z, \quad \tilde{\mathbf{B}}_0 = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \tag{135}$$

and we substitute this into the ansatz:

$$\begin{cases} \tilde{\mathbf{E}} = \underbrace{\hat{\mathbf{x}}E_x e^{i(kz-\omega t)}}_{\tilde{\mathbf{E}}_x} + \hat{\mathbf{y}}E_y e^{i(kz-\omega t)} + \hat{\mathbf{z}}E_z e^{i(kz-\omega t)}, \\ \tilde{\mathbf{B}} = \hat{\mathbf{x}}B_x e^{i(kz-\omega t)} + \hat{\mathbf{y}}B_y e^{i(kz-\omega t)} + \hat{\mathbf{z}}B_z e^{i(kz-\omega t)}. \end{cases} \tag{136}$$

The guided wave must satisfy:

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \boldsymbol{\nabla} \times \mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}.$$

We respectively deal with these shit:

1. $\boldsymbol{\nabla} \times \mathbf{E}$: Faraday's law gives:

$$\boldsymbol{\nabla} \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ \tilde{\mathbf{E}}_x & \tilde{\mathbf{E}}_y & \tilde{\mathbf{E}}_z \end{vmatrix} = \hat{\mathbf{x}}\left(\frac{\partial \tilde{\mathbf{E}}_z}{\partial y} - \frac{\partial \tilde{\mathbf{E}}_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial \tilde{\mathbf{E}}_x}{\partial z} - \frac{\partial \tilde{\mathbf{E}}_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial \tilde{\mathbf{E}}_y}{\partial x} - \frac{\partial \tilde{\mathbf{E}}_x}{\partial y}\right)$$

$$= \hat{\mathbf{x}}\left(\frac{\partial E_z}{\partial y}e^{i(\cdots)} - ikE_y e^{i(\cdots)}\right) + \hat{\mathbf{y}}\left(ikE_x e^{i(\cdots)} - \frac{\partial E_z}{\partial x}e^{i(\cdots)}\right) + \hat{\mathbf{z}}\left(\frac{\partial E_y}{\partial x}e^{i(\cdots)} - \frac{\partial E_x}{\partial y}e^{i(\cdots)}\right).$$

[7] $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ are arbitrary orthogonal basis due to the symmetry property.

Comparing this with the derivative of $\mathbf{B}$ with respect to t:

$$-\frac{\partial \mathbf{B}}{\partial t} = \hat{\mathbf{x}}\Big(i\omega B_x e^{i(\cdots)}\Big) + \hat{\mathbf{y}}\Big(i\omega B_y e^{i(\cdots)}\Big) + \hat{\mathbf{z}}\Big(i\omega B_z e^{i(\cdots)}\Big),$$

by canceling the exponential terms $e^{i(\cdots)}$, we have:

$$\begin{cases} \dfrac{\partial E_z}{\partial y} - ikE_y = i\omega B_x \\ ikE_x - \dfrac{\partial E_z}{\partial x} = i\omega B_y \\ \dfrac{\partial E_y}{\partial x} - \dfrac{\partial E_x}{\partial y} = i\omega B_z \end{cases} \tag{137}$$

2. $\boldsymbol{\nabla} \times \mathbf{B}$:Ampère's law gives that:

$$\boldsymbol{\nabla} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ \tilde{\mathbf{B}}_x & \tilde{\mathbf{B}}_y & \tilde{\mathbf{B}}_z \end{vmatrix} = \hat{\mathbf{x}}\left(\frac{\partial \tilde{\mathbf{B}}_z}{\partial y} - \frac{\partial \tilde{\mathbf{B}}_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial \tilde{\mathbf{B}}_x}{\partial z} - \frac{\partial \tilde{\mathbf{B}}_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial \tilde{\mathbf{B}}_y}{\partial x} - \frac{\partial \tilde{\mathbf{B}}_x}{\partial y}\right)$$

$$= \hat{\mathbf{x}}\left(\frac{\partial B_z}{\partial y}e^{i(\cdots)} - ikB_y e^{i(\cdots)}\right) + \hat{\mathbf{y}}\left(ikB_x e^{i(\cdots)} - \frac{\partial B_z}{\partial x}e^{i(\cdots)}\right) + \hat{\mathbf{z}}\left(\frac{\partial B_y}{\partial x}e^{i(\cdots)} - \frac{\partial B_x}{\partial y}e^{i(\cdots)}\right).$$

Comparing this with the derivative of $\mathbf{E}$ with respect to t:

$$\frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} = \hat{\mathbf{x}}\Big(-\frac{i\omega}{c^2}E_x e^{i(\cdots)}\Big) + \hat{\mathbf{y}}\Big(-\frac{i\omega}{c^2}E_y e^{i(\cdots)}\Big) + \hat{\mathbf{z}}\Big(-\frac{i\omega}{c^2}E_z e^{i(\cdots)}\Big),$$

by canceling the exponential terms $e^{i(\cdots)}$, we have:

$$\begin{cases} \dfrac{\partial B_z}{\partial y} - ikB_y = -\dfrac{i\omega}{c^2}E_x \\ ikB_x - \dfrac{\partial B_z}{\partial x} = -\dfrac{i\omega}{c^2}E_y \\ \dfrac{\partial B_y}{\partial x} - \dfrac{\partial B_x}{\partial y} = -\dfrac{i\omega}{c^2}E_z \end{cases} \tag{138}$$

Therefore, we have the three linear equation from (137) & (138) to solve, that is:

$$\begin{cases} \text{(i)}\ \dfrac{\partial E_z}{\partial y} - ikE_y = i\omega B_x \\ \text{(ii)}\ ikE_x - \dfrac{\partial E_z}{\partial x} = i\omega B_y \\ \text{(iii)}\ \dfrac{\partial E_y}{\partial x} - \dfrac{\partial E_x}{\partial y} = i\omega B_z \end{cases} \qquad \begin{cases} \text{(iv)}\ \dfrac{\partial B_z}{\partial y} - ikB_y = -\dfrac{i\omega}{c^2}E_x \\ \text{(v)}\ ikB_x - \dfrac{\partial B_z}{\partial x} = -\dfrac{i\omega}{c^2}E_y \\ \text{(vi)}\ \dfrac{\partial B_y}{\partial x} - \dfrac{\partial B_x}{\partial y} = -\dfrac{i\omega}{c^2}E_z \end{cases} \tag{139}$$

We can express E_x, E_y, B_x, B_y in terms of E_z and B_z. By (ii)+(iv):

$$iE_x\left(k+\frac{\omega}{c^2}\right)+\left(\frac{\partial B_z}{\partial y}-\frac{\partial E_z}{\partial x}\right)=i(k+\omega)B_y$$

$$\Rightarrow B_y=\left[\frac{k+(\omega/c^2)}{k+\omega}\right]E_x+\frac{1}{i(k+\omega)}\left(\frac{\partial B_z}{\partial y}-\frac{\partial E_z}{\partial x}\right),$$

Substituting B_y into (ii), we obtain

$$ikE_x-\frac{\partial E_z}{\partial x}=i\omega\left[\frac{k+(\omega/c^2)}{k+\omega}\right]E_x+\frac{\omega}{k+\omega}\left(\frac{\partial B_z}{\partial y}-\frac{\partial E_z}{\partial x}\right)$$

$$\Rightarrow iE_x\left[k-\frac{k\omega+(\omega^2/c^2)}{k+\omega}\right]=i\left[\frac{k^2-(\omega^2/c^2)}{k+\omega}\right]E_x=\frac{\partial E_z}{\partial x}+\frac{\omega}{k+\omega}\left(\frac{\partial B_z}{\partial y}-\frac{\partial E_z}{\partial x}\right),$$

which leads to:

$$\boxed{E_x=\frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial E_z}{\partial x}+\omega\frac{\partial B_z}{\partial y}\right).} \tag{140}$$

Substituting (140) into (iv):

$$\frac{\partial B_z}{\partial y}-ikB_y=-\frac{i\omega}{c^2}\left[\frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial E_z}{\partial x}+\omega\frac{\partial B_z}{\partial y}\right)\right]$$

$$\Rightarrow ikB_y=\frac{\partial B_z}{\partial y}-\frac{\omega}{(\omega/c)^2-k^2}\left(\frac{k}{c^2}\frac{\partial E_z}{\partial x}+\frac{\omega}{c^2}\frac{\partial B_z}{\partial y}\right)=-\frac{k\omega}{(\omega/c)^2-k^2}\frac{1}{c^2}\frac{\partial E_z}{\partial x}-\frac{k^2}{(\omega/c)^2-k^2}\frac{\partial B_z}{\partial y},$$

dividing both sides by ik, and we can obtain

$$\boxed{B_y=\frac{i}{(\omega/c)^2-k^2}\left(k\frac{\partial B_z}{\partial y}+\frac{\omega}{c^2}\frac{\partial E_z}{\partial x}\right).} \tag{141}$$

Similarly, (i)+(v) gives that

$$i(k-\omega)B_x=i\left(k-\frac{\omega}{c^2}\right)E_y-\left(\frac{\partial E_z}{\partial y}-\frac{\partial B_z}{\partial x}\right)$$

$$\Rightarrow B_x=\frac{k-(\omega/c^2)}{k-\omega}E_y-\frac{i}{\omega-k}\left(\frac{\partial E_z}{\partial y}-\frac{\partial B_z}{\partial x}\right)$$

Substituting B_x into (v), we have:

$$ik\left[\frac{k-(\omega/c^2)}{k-\omega}E_y-\frac{i}{\omega-k}\left(\frac{\partial E_z}{\partial y}-\frac{\partial B_z}{\partial x}\right)\right]-\frac{\partial B_z}{\partial x}=-\frac{i\omega}{c^2}E_y$$

$$\Rightarrow iE_y\left[\frac{k^2-(\omega/c)^2}{k-\omega}\right]=\frac{\partial B_z}{\partial x}+\frac{k}{k-\omega}\left(\frac{\partial E_z}{\partial y}-\frac{\partial B_z}{\partial x}\right)=\frac{1}{k-\omega}\left(k\frac{\partial E_z}{\partial y}-\omega\frac{\partial B_z}{\partial x}\right),$$

thus we can get

$$\boxed{E_y = \frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial E_z}{\partial y} - \omega\frac{\partial B_z}{\partial x}\right).} \tag{142}$$

Substituting E_y into (i):

$$\frac{\partial E_z}{\partial y} - ik\left[\frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial E_z}{\partial y} - \omega\frac{\partial B_z}{\partial x}\right)\right] = i\omega B_x$$

$$\Rightarrow \frac{\partial E_z}{\partial y} + \frac{k}{(\omega/c)^2 - k^2}\left(k\frac{\partial E_z}{\partial y} - \omega\frac{\partial B_z}{\partial x}\right) = \frac{(\omega/c)^2}{(\omega/c)^2 - k^2}\frac{\partial E_z}{\partial y} - \frac{k\omega}{(\omega/c)^2 - k^2}\frac{\partial B_z}{\partial x} = i\omega B_x,$$

and divide both sides by $i\omega$, we can get the final part of the x, y components:

$$\boxed{B_x = \frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial B_z}{\partial x} - \frac{\omega}{c^2}\frac{\partial E_z}{\partial y}\right).} \tag{143}$$

All the solutions are:

$$\begin{aligned}
&\text{Electric}\begin{cases} E_x = \dfrac{i}{(\omega/c)^2 - k^2}\left(k\dfrac{\partial E_z}{\partial x} + \omega\dfrac{\partial B_z}{\partial y}\right) \\ E_y = \dfrac{i}{(\omega/c)^2 - k^2}\left(k\dfrac{\partial E_z}{\partial y} - \omega\dfrac{\partial B_z}{\partial x}\right) \end{cases} \\
&\text{Magnetic}\begin{cases} B_x = \dfrac{i}{(\omega/c)^2 - k^2}\left(k\dfrac{\partial B_z}{\partial x} - \dfrac{\omega}{c^2}\dfrac{\partial E_z}{\partial y}\right) \\ B_y = \dfrac{i}{(\omega/c)^2 - k^2}\left(k\dfrac{\partial B_z}{\partial y} + \dfrac{\omega}{c^2}\dfrac{\partial E_z}{\partial x}\right). \end{cases}
\end{aligned} \tag{144}$$

These equation and expression are written in terms of E_z and B_z, namely, the longitudinal components. By Gauss's law $\boldsymbol{\nabla} \cdot \tilde{\mathbf{E}} = 0$ and $\boldsymbol{\nabla} \cdot \tilde{\mathbf{B}} = 0$, we can write down:

$$\boldsymbol{\nabla} \cdot \tilde{\mathbf{E}} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0, \quad \boldsymbol{\nabla} \cdot \tilde{\mathbf{B}} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0,$$

with

$$\frac{\partial E_z}{\partial z} = ikE_z, \quad \frac{\partial B_z}{\partial z} = ikB_z. \tag{145}$$

Substituting (144) into the divergence, we have:

$$\begin{cases} \dfrac{\partial E_x}{\partial x} = \dfrac{ik}{(\omega/c)^2 - k^2}\dfrac{\partial^2 E_z}{\partial x^2} + \dfrac{i\omega}{(\omega/c)^2 - k^2}\dfrac{\partial^2 B_z}{\partial x \partial y} \\ \dfrac{\partial E_y}{\partial y} = \dfrac{ik}{(\omega/c)^2 - k^2}\dfrac{\partial^2 E_z}{\partial y^2} - \dfrac{i\omega}{(\omega/c)^2 - k^2}\dfrac{\partial^2 B_z}{\partial x \partial y} \end{cases}$$

$$\begin{cases} \dfrac{\partial B_x}{\partial x} = \dfrac{ik}{(\omega/c)^2 - k^2}\dfrac{\partial^2 B_z}{\partial x^2} - \dfrac{i(\omega/c^2)}{(\omega/c)^2 - k^2}\dfrac{\partial^2 E_z}{\partial x \partial y} \\ \dfrac{\partial B_y}{\partial y} = \dfrac{ik}{(\omega/c)^2 - k^2}\dfrac{\partial^2 B_z}{\partial y^2} + \dfrac{i(\omega/c^2)}{(\omega/c)^2 - k^2}\dfrac{\partial^2 E_z}{\partial x \partial y}. \end{cases}$$

Therefore, the divergence gives that

$$\partial_x E_x + \partial_y E_y + \partial_z E_z = (\partial_x E_x + \partial_y E_y) + ikE_z = \frac{ik}{(\omega/c)^2 - k^2}(\partial_x^2 + \partial_y^2)E_z + ikE_z = 0$$

Then we can use the equation to decouple the $z-$component, and turn it into a linear equation. So does B_z, which are:

$$\begin{cases} \left[\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] E_z = 0 \\ \left[\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] B_z = 0, \end{cases} \tag{146}$$

where $\partial_x^2 + \partial_y^2 + (\omega/c)^2 - k^2$ is a linear differential operator.

1. **When** $\underline{E_z = 0}$: It's called **TE** wave, which means *transverse electric*. The electric field propagates along $\hat{\mathbf{z}}$ and just oscillates on xy plane.

2. **When** $\underline{B_z = 0}$: It's called **TM** wave, which means *transverse magnetic*. The magnetic field propagates along $\hat{\mathbf{z}}$ and just oscillates on xy plane.

3. **When** $\underline{E_z = 0}$ **and** $\underline{B_z = 0}$: It's called **TEM** wave, but in this case, it's impossible to have wave in the wave guide. Since

$$\begin{cases} \dfrac{\partial E_x}{\partial x} + \dfrac{\partial E_y}{\partial y} = 0 & \text{Gauss's Law} \\ \dfrac{\partial E_y}{\partial x} - \dfrac{\partial E_x}{\partial y} = -\dfrac{\partial B_z}{\partial t} = 0 & \text{Faraday's Law,} \end{cases} \tag{147}$$

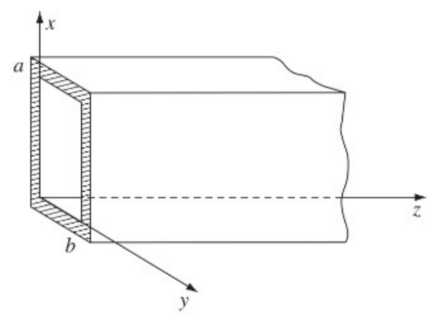
so, the electric field $\tilde{\mathbf{E}}$ is curlless and divergenceless. It implies that we can let $\tilde{\mathbf{E}} = \boldsymbol{\nabla}\tilde{V}$, and Gauss's Law gives

$$\nabla^2 \tilde{V} = 0, \tag{148}$$

which is a Laplace equation. Laplace equation tells us that within the boundary, **there's no local extremum**. By the boundary conditions (133), $\mathbf{E} = \mathbf{0}$ on the boundary, so $\mathbf{E} = 0$ everywhere in the wave guide, which means that there's no electric wave and therefore no magnetic wave.

6.2 Rectangular Case

Consider a **TE** wave propagates along $\hat{\mathbf{z}}$ in a rectangular wave guide, as shown below:



So, in this case,

$$E_z = 0, \quad B_z = B_{z_0}(x, y)e^{i(kz-\omega t)}. \tag{149}$$

By separation of variables, let

$$B_z = X(x)Y(y)e^{i(kz-\omega t)}, \tag{150}$$

substitute this into (146), we can write

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] X(x)Y(y)e^{i(kz-\omega t)} = \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] X(x)Y(y) = 0. \tag{151}$$

Because the differential operator is independent of z, t, we can drop the exponential term. In addition, the boundary conditions (133) indicate that

$$\begin{cases} x = 0, a : \quad \underbrace{E_y = 0}_{\text{Parallel } \mathbf{E}^{\parallel} = \mathbf{0}} \quad , \qquad \underbrace{B_x = 0}_{\text{Perpendicular } B^{\perp} = 0} \\ y = 0, b : E_x = 0, \quad B_y = 0. \end{cases} \tag{152}$$

Assume that

$$X''(x) = -k_x^2 X(x), \quad Y''(y) = -k_y^2 Y(y), \tag{153}$$

and the boundary conditions give:

$$\begin{cases} X'(x)\Big|_{x=a,0} = 0 \Rightarrow X(x) = * \cos\left(\frac{m\pi x}{a}\right) \\ Y'(y)\Big|_{y=b,0} = 0 \Rightarrow Y(y) = * \cos\left(\frac{n\pi y}{b}\right). \end{cases} \tag{154}$$

Note that $*$ means the constant which will later be absorbed into b_{mn}. That is, the wave equation (151) gives that:

$$
\begin{aligned}
&-\left(\frac{m\pi}{a}\right)^2 \cos\cos - \left(\frac{n\pi}{b}\right)^2 \cos\cos + \left(\frac{\omega}{c}\right)^2 \cos\cos - k^2 \cos\cos = 0 \\
&\Rightarrow k^2 = \left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n}{b}\right)^2 \equiv k_{mn} = \pm\sqrt{\left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n}{b}\right)^2}.
\end{aligned} \tag{155}
$$

Hence, the general solution to a given frequency ω is:

$$
B_z = \sum_{m,n=0,1,\cdots} \left[b_{mn} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{i(k_{mn}z-\omega t)} \right]. \tag{156}
$$

If there's no given ω, the constant b_{mn} and k_{mn} are the function of ω, the general solution becomes

$$
B_z = \sum_{m,n=0,1,\cdots} \int \mathrm{d}\omega\, b_{mn}(\omega) \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \exp i\Big(k_{mn}(\omega)z - \omega t\Big). \tag{157}
$$

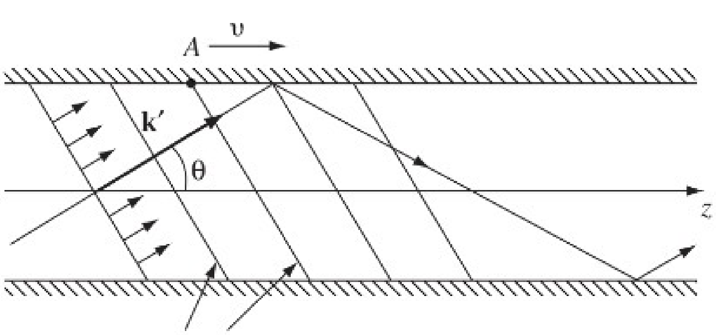
Note that the wave numbers for x and y are $m\pi/a$ and $n\pi/b$, so the wave vector is

$$
\mathbf{k} = \pm\hat{\mathbf{x}}\frac{m\pi}{a} \pm \hat{\mathbf{y}}\frac{n\pi}{b} + \hat{\mathbf{z}}k_{mn}. \tag{158}
$$

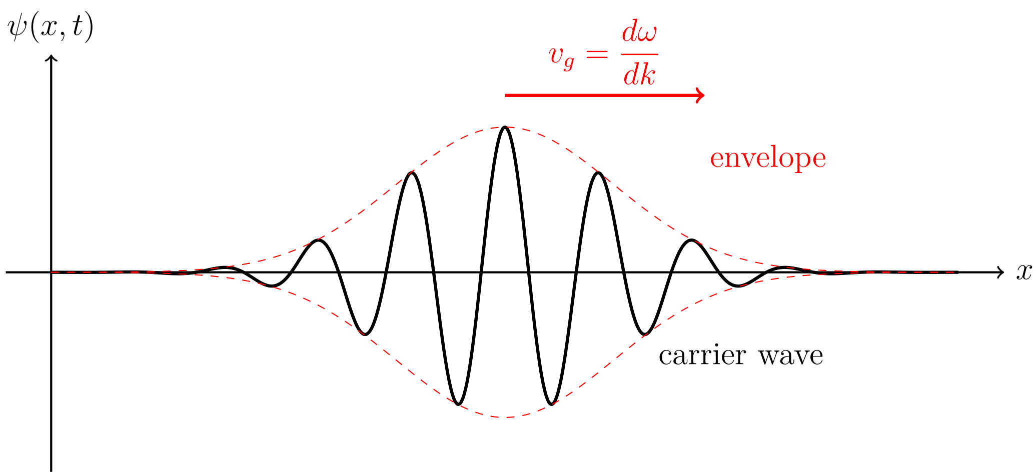
The **Phase velocity** v in this case is:

$$
v = \frac{\omega}{k_{mn}} = \frac{c}{\sqrt{1 - \left(\frac{c}{\omega}\right)^2 \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]}} > c, \tag{159}
$$

which is faster than the speed of light, that's because the projection of the wave front makes the wavelength larger, so the speed of the propagation of phase is faster than that of true wave. As shown below:



In fact, the energy is carried by the **Group velocity**, the illustration is shown below:



The group velocity is defined as

$$v_g \equiv \frac{d\omega}{dk}, \tag{160}$$

which can also be denoted by:

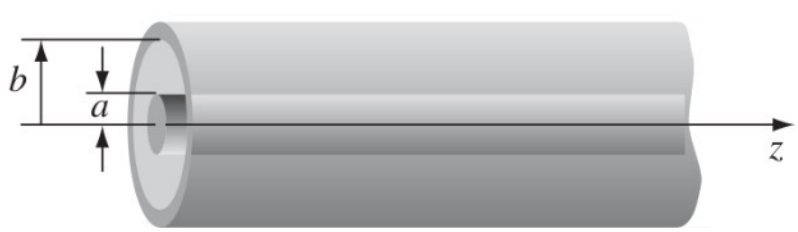
$$v_g = \left(\frac{dk(\omega)}{d\omega}\right)^{-1} = \left[\frac{\omega/c^2}{\dfrac{\omega}{c}\sqrt{1-\left(\dfrac{c}{\omega}\right)^2\left[\left(\dfrac{m\pi}{a}\right)^2+\left(\dfrac{n\pi}{b}\right)^2\right]}}\right]^{-1} = \left[\frac{1}{c\sqrt{1-\left(\dfrac{c}{\omega}\right)^2\left[\left(\dfrac{m\pi}{a}\right)^2+\left(\dfrac{n\pi}{b}\right)^2\right]}}\right]^{-1},$$

hence, the group velocity is

$$v_g = c\sqrt{1-\left(\frac{c}{\omega}\right)^2\left[\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]} < c. \tag{161}$$

6.3 Coaxial Case

The hollow wave guide cannot support TEM wave, but a **coaxial transmission line can**. As shown below:



In this configuration, let $E_z, B_z = 0$, and with $k = \omega/c$. By (139), the propagation gives that

$$\begin{cases} \dfrac{\partial E_z}{\partial y} - ikE_y = -ikE_y = i\omega B_x \Rightarrow cB_x = -E_y \\ \dfrac{\partial B_z}{\partial y} - ikB_y = -ikB_y = -\dfrac{i\omega}{c^2}E_x \Rightarrow cB_y = E_x, \end{cases} \tag{162}$$

this shows that $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to each other with TEM wave. Keep going, (139) gives that:

$$\begin{cases} \dfrac{\partial E_x}{\partial x} + \dfrac{\partial E_y}{\partial y} = 0 \Rightarrow \boldsymbol{\nabla} \cdot \mathbf{E} = 0 \\ \dfrac{\partial E_y}{\partial x} - \dfrac{\partial E_x}{\partial y} = 0 \Rightarrow \boldsymbol{\nabla} \times \mathbf{E} = \mathbf{0} \\ \dfrac{\partial B_x}{\partial x} + \dfrac{\partial B_y}{\partial y} = 0 \Rightarrow \boldsymbol{\nabla} \cdot \mathbf{B} = 0 \\ \dfrac{\partial B_y}{\partial x} - \dfrac{\partial B_x}{\partial y} = 0 \Rightarrow \boldsymbol{\nabla} \times \mathbf{B} = \mathbf{0}. \end{cases} \tag{163}$$

Note that the equations in (163) are the conditions of **electrostatics** and **magnetostatics**! Therefore, we can use the method before to determine the electric field and the magnetic field:

$$\begin{aligned} &\oint \mathrm{d}\mathbf{a} \cdot \mathbf{E} = E \cdot 2\pi s l = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{\pi a^2 l \rho}{\epsilon_0} \Rightarrow \mathbf{E} = \frac{\rho a^2}{2\epsilon_0}\frac{1}{s}\hat{\mathbf{s}} \equiv \frac{A}{s}\hat{\mathbf{s}}, \\ &\oint \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{B} = B \cdot 2\pi s = \mu_0 I \Rightarrow \mathbf{B} = \frac{\mu_0 I}{2\pi}\frac{1}{s}\hat{\boldsymbol{\phi}} = \frac{A}{cs}\hat{\boldsymbol{\phi}}, \end{aligned} \tag{164}$$

where we can always work out to make the same A constant, with two equations being only off by a constant (light speed c) factor.
Ansatz: The wave function is

$$\tilde{\mathbf{E}}(s, \phi, z, t) = \tilde{\mathbf{E}}_0 e^{i(kz-\omega t)}, \quad \tilde{\mathbf{B}}(s, \phi, z, t) = \tilde{\mathbf{B}}_0 e^{i(kz-\omega t)}. \tag{165}$$

Substituting (164) into the ansatz and take the real part(In this case, we explicitly determined the fields. So it's easy to do so.), we obtain:

$$\begin{cases} \mathbf{E}(s, \phi, z, t) = \dfrac{A\cos(kz - \omega t)}{s}\hat{\mathbf{s}} \\ \mathbf{B}(s, \phi, z, t) = \dfrac{A\cos(kz - \omega t)}{cs}\hat{\boldsymbol{\phi}}. \end{cases} \tag{166}$$

Appendix: Guided Wave in Cylindrical Coordinate

General Solution to Bessel Equation

The Bessel equation is given by:

$$x^2y'' + xy' + (a^2x^2 - p^2)y = 0. \tag{167}$$

The general solution to this shit is called **Bessel Function**:

$$y = AJ_p(ax) + BY_p(ax), \tag{168}$$

where Y_p is also called **Neumann Function**. While $x = 0$, Y_p will diverge[a], so if we want the solution to converge, we need to drop Y_p.
If $ax = x_{mn}$, then

$$J_m(x_{mn}) = 0. \tag{169}$$

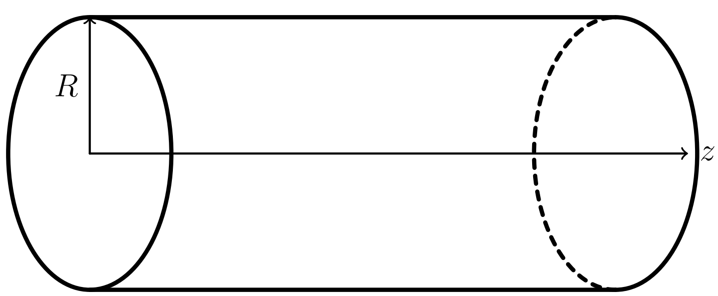
Then the discrete general solution becomes:

$$y = \sum_{\substack{m=0,1,\cdots \\ n=1,2,\cdots}} A_{mn}J_m(x_{mn}x)(\text{other term}). \tag{170}$$

[a] $Y_p \sim x^{-p}$, so if $x = 0$, it diverges.

Cylindrical Wave Guide

Consider a cylindrical wave guide down the $z-$axis. As shown below.



In this case, we need to change the coordinates from Cartesian to Cylindrical. Of course, we still need to consider the boundary conditions:

$$\begin{cases} \mathbf{E}^{\parallel} = \mathbf{0} \\ \\ B^{\perp} = 0 \end{cases}$$

So, for electric field, E_z and E_ϕ is 0 on the boundary since they are parallel to the boundary. Furthermore, for magnetic field, B_s is zero, trivial.

Ansatz: The waves are now

$$\begin{cases} \tilde{\mathbf{E}}(s,\phi,z,t) = \tilde{\mathbf{E}}_0(s,\phi)e^{i(kz-\omega t)}, \\ \tilde{\mathbf{B}}(s,\phi,z,t) = \tilde{\mathbf{B}}_0(s,\phi)e^{i(kz-\omega)}. \end{cases} \tag{171}$$

However, the waves are not always transverse(just along $\hat{\mathbf{s}}$ or $\hat{\boldsymbol{\phi}}$) since it's strictly constrained in the wave guide made of a perfect conductor. So, it may have z-component:

$$\tilde{\mathbf{E}}_0 = \hat{\mathbf{s}}E_s + \hat{\boldsymbol{\phi}}E_\phi + \hat{\mathbf{z}}E_z, \quad \tilde{\mathbf{B}}_0 = \hat{\mathbf{s}}B_s + \hat{\boldsymbol{\phi}}B_\phi + \hat{\mathbf{z}}B_z. \tag{172}$$

The guided wave must satisfy:

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \boldsymbol{\nabla} \times \mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}.$$

We respectively deal with these shit:

1. $\boldsymbol{\nabla} \times \mathbf{E}$: Faraday's law gives:

$$\boldsymbol{\nabla} \times \mathbf{E} = e^{i(\cdots)}\left[\hat{\mathbf{s}}\left(\frac{1}{s}\frac{\partial E_z}{\partial \phi} - \frac{\partial E_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial E_s}{\partial z} - \frac{\partial E_z}{\partial s}\right) + \hat{\mathbf{z}}\frac{1}{s}\left(\frac{\partial (sE_\phi)}{\partial s} - \frac{\partial E_s}{\partial \phi}\right)\right].$$

Comparing this with the derivative of $\mathbf{B}$ with respect to t:

$$-\frac{\partial \mathbf{B}}{\partial t} = e^{i(\cdots)}\left[\hat{\mathbf{s}}\Big(i\omega B_s\Big) + \hat{\mathbf{y}}\Big(i\omega B_\phi\Big) + \hat{\mathbf{z}}\Big(i\omega B_z\Big)\right],$$

by canceling the exponential terms $e^{i(\cdots)}$, we have:

$$\begin{cases} \dfrac{1}{s}\dfrac{\partial E_z}{\partial \phi} - ikE_\phi = i\omega B_s \\ ikE_s - \dfrac{\partial E_z}{\partial s} = i\omega B_\phi \\ \dfrac{1}{s}\left(\dfrac{\partial (sE_\phi)}{\partial s} - \dfrac{\partial E_s}{\partial \phi}\right) = i\omega B_z \end{cases} \tag{173}$$

2. $\boldsymbol{\nabla} \times \mathbf{B}$: Ampère's law gives the similar result:

$$\begin{cases} \dfrac{1}{s}\dfrac{\partial B_z}{\partial \phi} - ikB_\phi = -\dfrac{i\omega}{c^2}E \\ ikB_s - \dfrac{\partial B_z}{\partial s} = -\dfrac{i\omega}{c^2}E_\phi \\ \dfrac{1}{s}\left(\dfrac{\partial (sB_\phi)}{\partial s} - \dfrac{\partial B_s}{\partial \phi}\right) = -\dfrac{i\omega}{c^2}E_z \end{cases} \tag{174}$$

Therefore, we have the three linear equation from (173) & (174) to solve, that is:

$$\begin{cases} \text{(i)}\ \dfrac{1}{s}\dfrac{\partial E_z}{\partial \phi} - ikE_\phi = i\omega B_s \\ \text{(ii)}\ ikE_s - \dfrac{\partial E_z}{\partial s} = i\omega B_\phi \\ \text{(iii)}\ \dfrac{1}{s}\left(\dfrac{\partial (sE_\phi)}{\partial s} - \dfrac{\partial E_s}{\partial \phi}\right) = i\omega B_z \end{cases} \qquad \begin{cases} \text{(iv)}\ \dfrac{1}{s}\dfrac{\partial B_z}{\partial \phi} - ikB_\phi = -\dfrac{i\omega}{c^2}E_s \\ \text{(v)}\ ikB_s - \dfrac{\partial B_z}{\partial s} = -\dfrac{i\omega}{c^2}E_\phi \\ \text{(vi)}\ \dfrac{1}{s}\left(\dfrac{\partial (sB_\phi)}{\partial s} - \dfrac{\partial B_s}{\partial \phi}\right) = -\dfrac{i\omega}{c^2}E_z \end{cases} \tag{175}$$

We can express E_s, E_ϕ, B_s, B_ϕ in terms of E_z and B_z. By (ii)+(iv):

$$iE_s\left(k + \frac{\omega}{c^2}\right) + \left(\frac{1}{s}\frac{\partial B_z}{\partial \phi} - \frac{\partial E_z}{\partial s}\right) = i(k+\omega)B_\phi$$

$$\Rightarrow B_\phi = \left[\frac{k + (\omega/c^2)}{k+\omega}\right]E_s + \frac{1}{i(k+\omega)}\left(\frac{1}{s}\frac{\partial B_z}{\partial \phi} - \frac{\partial E_z}{\partial s}\right),$$

Substituting B_ϕ into (ii), and we define $k_c^2 = (\omega/c)^2 - k^2$, we obtain

$$ikE_s - \frac{\partial E_z}{\partial s} = i\omega\left[\frac{k + (\omega/c)^2}{k+\omega}\right]E_s + \frac{\omega}{k+\omega}\left(\frac{1}{s}\frac{\partial B_z}{\partial \phi} - \frac{\partial E_z}{\partial s}\right)$$

$$\Rightarrow i\left[\frac{-k_c^2}{k+\omega}\right]E_s = \frac{\partial E_z}{\partial s} + \frac{\omega}{k+\omega}\left(\frac{1}{s}\frac{\partial B_z}{\partial \phi} - \frac{\partial E_z}{\partial s}\right),$$

which leads to:

$$\boxed{E_s = \frac{i}{k_c^2}\left(k\frac{\partial E_z}{\partial s} + \frac{\omega}{s}\frac{\partial B_z}{\partial \phi}\right).} \tag{176}$$

Substituting (176) into (iv):

$$\frac{1}{s}\frac{\partial B_z}{\partial \phi} - ikB_\phi = -\frac{i\omega}{c^2}\left[\frac{i}{k_c^2}\left(k\frac{\partial E_z}{\partial s} + \frac{\omega}{s}\frac{\partial B_z}{\partial \phi}\right)\right]$$

$$\Rightarrow ikB_\phi = \frac{1}{s}\frac{\partial B_z}{\partial \phi} - \frac{\omega}{k_c^2}\left(\frac{k}{c^2}\frac{\partial E_z}{\partial s} + \frac{\omega}{sc^2}\frac{\partial B_z}{\partial \phi}\right) = -\frac{k\omega}{k_c^2}\frac{1}{c^2}\frac{\partial E_z}{\partial s} - \frac{k^2}{k_c^2}\frac{1}{s}\frac{\partial B_z}{\partial \phi},$$

dividing both sides by ik, and we can obtain

$$\boxed{B_\phi = \frac{i}{k_c^2}\left(k\frac{1}{s}\frac{\partial B_z}{\partial \phi} + \frac{\omega}{c^2}\frac{\partial E_z}{\partial s}\right).} \tag{177}$$

Similarly, (i)+(v) gives that

$$i(k-\omega)B_s = i\left(k - \frac{\omega}{c^2}\right)E_\phi - \left(\frac{1}{s}\frac{\partial E_z}{\partial \phi} - \frac{\partial B_z}{\partial s}\right)$$

$$\Rightarrow B_s = \frac{k-(\omega/c^2)}{k-\omega}E_\phi + \frac{i}{k-\omega}\left(\frac{1}{s}\frac{\partial E_z}{\partial \phi} - \frac{\partial B_z}{\partial s}\right)$$

Substituting B_s into (v), we have:

$$ik\left[\frac{k-(\omega/c^2)}{k-\omega}E_\phi + \frac{i}{k-\omega}\left(\frac{1}{s}\frac{\partial E_z}{\partial \phi} - \frac{\partial B_z}{\partial s}\right)\right] - \frac{\partial B_z}{\partial s} = -\frac{i\omega}{c^2}E_\phi$$

$$\Rightarrow iE_\phi\left[\frac{-k_c^2}{k-\omega}\right] = \frac{\partial B_z}{\partial s} + \frac{k}{k-\omega}\left(\frac{1}{s}\frac{\partial E_z}{\partial \phi} - \frac{\partial B_z}{\partial s}\right) = \frac{1}{k-\omega}\left(\frac{k}{s}\frac{\partial E_z}{\partial \phi} - \omega\frac{\partial B_z}{\partial x}\right),$$

thus we can get

$$\boxed{E_\phi = \frac{i}{k_c^2}\left(\frac{k}{s}\frac{\partial E_z}{\partial \phi} - \omega\frac{\partial B_z}{\partial x}\right).} \tag{178}$$

Substituting E_ϕ into (i):

$$\frac{1}{s}\frac{\partial E_z}{\partial \phi} - ik\left[\frac{i}{k_c^2}\left(\frac{k}{s}\frac{\partial E_z}{\partial \phi} - \omega\frac{\partial B_z}{\partial s}\right)\right] = i\omega B_s$$

$$\Rightarrow \frac{(\omega/c)^2}{k_c^2}\frac{1}{s}\frac{\partial E_z}{\partial \phi} - \frac{\omega k}{k_c^2}\frac{\partial B_z}{\partial s} = i\omega B_s,$$

and divide both sides by $i\omega$, we can get the final part of the s, ϕ components:

$$\boxed{B_s = \frac{i}{k_c^2}\left(k\frac{\partial B_z}{\partial s} - \frac{\omega}{c^2}\frac{1}{s}\frac{\partial E_z}{\partial \phi}\right).} \tag{179}$$

All the solutions are:

$$\begin{aligned}
&\text{Electric}\begin{cases} E_s = \dfrac{i}{k_c^2}\left(k\dfrac{\partial E_z}{\partial s} + \dfrac{\omega}{s}\dfrac{\partial B_z}{\partial \phi}\right) \\[2ex] E_\phi = \dfrac{i}{k_c^2}\left(\dfrac{k}{s}\dfrac{\partial E_z}{\partial \phi} - \omega\dfrac{\partial B_z}{\partial s}\right) \end{cases} \\
&\text{Magnetic}\begin{cases} B_s = \dfrac{i}{k_c^2}\left(k\dfrac{\partial B_z}{\partial s} - \dfrac{\omega}{c^2}\dfrac{1}{s}\dfrac{\partial E_z}{\partial \phi}\right) \\[2ex] B_\phi = \dfrac{i}{k_c^2}\left(k\dfrac{1}{s}\dfrac{\partial B_z}{\partial \phi} + \dfrac{\omega}{c^2}\dfrac{\partial E_z}{\partial s}\right). \end{cases}
\end{aligned} \tag{180}$$

These equations and expressions are written in terms of E_z and B_z, namely, the longitudinal components. By Gauss's law $\boldsymbol{\nabla}\cdot\tilde{\mathbf{E}}=0$ and $\boldsymbol{\nabla}\cdot\tilde{\mathbf{B}}=0$, we can write down:

$$\boldsymbol{\nabla}\cdot\tilde{\mathbf{E}}=\frac{1}{s}\frac{\partial(sE_s)}{\partial s}+\frac{1}{s}\frac{\partial E_\phi}{\partial\phi}+\frac{\partial E_z}{\partial z}=0,\quad \boldsymbol{\nabla}\cdot\tilde{\mathbf{B}}=\frac{1}{s}\frac{\partial(sB_s)}{\partial s}+\frac{1}{s}\frac{\partial B_\phi}{\partial\phi}+\frac{\partial B_z}{\partial z}=0,$$

with

$$\frac{\partial E_z}{\partial z}=ikE_z,\quad \frac{\partial B_z}{\partial z}=ikB_z. \tag{181}$$

Substituting (180) into the divergence, we have:

$$\begin{cases}\dfrac{1}{s}\dfrac{\partial(sE_s)}{\partial s}=\dfrac{1}{s}\dfrac{ik}{k_c^2}\dfrac{\partial E_z}{\partial s}+\dfrac{ik}{k_c^2}\dfrac{\partial^2 E_z}{\partial s^2}+\dfrac{1}{s}\dfrac{i\omega}{k_c^2}\dfrac{\partial^2 B_z}{\partial s\partial\phi}\\[2ex] \dfrac{1}{s}\dfrac{\partial E_\phi}{\partial\phi}=\dfrac{1}{s^2}\dfrac{ik}{k_c^2}\dfrac{\partial^2 E_z}{\partial\phi^2}-\dfrac{1}{s}\dfrac{i\omega}{k_c^2}\dfrac{\partial^2 B_z}{\partial s\partial\phi}\end{cases}$$

$$\begin{cases}\dfrac{1}{s}\dfrac{\partial(sB_s)}{\partial s}=\dfrac{1}{s}\dfrac{ik}{k_c^2}\dfrac{\partial B_z}{\partial s}+\dfrac{ik}{k_c^2}\dfrac{\partial^2 B_z}{\partial s^2}-\dfrac{i\omega}{sc^2k_c^2}\dfrac{\partial^2 E_z}{\partial s\partial\phi}\\[2ex] \dfrac{1}{s}\dfrac{\partial B_\phi}{\partial\phi}=\dfrac{1}{s^2}\dfrac{ik}{k_c^2}\dfrac{\partial^2 B_z}{\partial\phi^2}+\dfrac{1}{s}\dfrac{i\omega}{k_c^2}\dfrac{\partial^2 E_z}{\partial s\partial\phi}\end{cases}$$

Therefore, the divergence gives that

$$\frac{1}{s}\partial_s(sE_s)+\frac{1}{s}\partial_\phi E_\phi+\partial_z E_z=\frac{ik}{k_c^2}\partial_s^2E_z+\frac{1}{s}\frac{ik}{k_c^2}\partial_sE_z+\frac{1}{s^2}\frac{ik}{k_c^2}\partial_\phi^2E_\phi+ikE_z=0.$$

Then we can use the equation to decouple the $z-$component, and turn it into a linear equation. So does B_z, which are:

$$\begin{cases}\left[\dfrac{\partial^2}{\partial s^2}+\dfrac{1}{s}\dfrac{\partial}{\partial s}+\dfrac{1}{s^2}\dfrac{\partial^2}{\partial\phi^2}+k_c^2\right]E_z=0\\[2ex] \left[\dfrac{\partial^2}{\partial s^2}+\dfrac{1}{s}\dfrac{\partial}{\partial s}+\dfrac{1}{s^2}\dfrac{\partial^2}{\partial\phi^2}+k_c^2\right]B_z=0,\end{cases} \tag{182}$$

so, we need to use (182) to solve E_z or B_z since they are decoupled terms. After obtaining z components, substituting the results into (180) and getting the other four terms.
Back to the cylindrical wave guide, we can consider a **TM Wave**: When a TM wave propagates down the $\hat{\mathbf{z}}$ in a cylindrical wave guide, then $B_z=0$ and we use the separation of variables:

$$E_z(s,\phi)=S(s)\Phi(\phi), \tag{183}$$

then substitute this into (182), we obtain a differential equation:

$$\left[\frac{\partial^2}{\partial s^2}+\frac{1}{s}\frac{\partial}{\partial s}+\frac{1}{s^2}\frac{\partial^2}{\partial\phi^2}+k_c^2\right]S(s)\Phi(\phi)=0. \tag{184}$$

Therefore, we can separate this into

$$\begin{cases} \text{(i) } s^2 S''(s) + sS'(s) + (k_c^2 s^2 - m^2)S(s) = 0 \\ \text{(i) } \dfrac{1}{\Phi}\Phi''(\phi) = -m^2, \end{cases} \tag{185}$$

and (ii) gives that

$$\Phi(\phi) = e^{-im\phi}. \tag{186}$$

Note that (i) is a Bessel equation, so the general solution to $S(s)$ is:

$$S(s) = AJ_m(k_c s) + BY_m(k_c s). \tag{187}$$

It's finite when $s = 0$, so Y_m can be dropped. With the boundary condition, when $s = R$, the electric field is 0, namely

$$S(R) = AJ_m(k_c R) = 0 \Rightarrow k_c = \frac{s_{mn}}{R},$$

therefore,

$$S(s) = AJ_m\left(\frac{s_{mn}}{R}s\right). \tag{188}$$

Substituting (186) and into (183), the z component of the electric field is

$$E_z = E_0 J_m\left(\frac{s_{mn}}{R}s\right)e^{im\phi}, \tag{189}$$

where E_0 is the amplitude of the field. As the result, substituting E_z into (180), we have

$$\begin{cases} E_s = \dfrac{ik}{k_c^2}\dfrac{\partial E_z}{\partial s} = \dfrac{ik}{(s_{mn}/R)^2}\dfrac{s_{mn}}{R}E_0 J_m'\left(\dfrac{s_{mn}}{R}s\right)e^{im\phi} = \dfrac{iRk}{s_{mn}}E_0 J_m'\left(\dfrac{s_{mn}}{R}s\right)e^{im\phi}, \\ E_\phi = \dfrac{ik}{k_c^2 s}\dfrac{\partial E_z}{\partial \phi} = -\dfrac{kmR^2}{s_{mn}^2 s}E_0 J_m\left(\dfrac{s_{mn}}{R}s\right)e^{im\phi}. \end{cases} \tag{190}$$

So the solution to the electric field $\tilde{\mathbf{E}}(s, \phi, z, t)$ is:

$$\boxed{\tilde{\mathbf{E}}_{mn} = E_0 e^{im\phi} e^{i(kz-\omega t)}\left[\hat{\mathbf{r}}\frac{iRk}{s_{mn}}J_m'\left(\frac{s_{mn}}{R}s\right) - \hat{\boldsymbol{\phi}}\frac{kmR^2}{s_{mn}^2 s}J_m\left(\frac{s_{mn}}{R}s\right) + \hat{\mathbf{z}}J_m\left(\frac{s_{mn}}{R}s\right)\right]}. \tag{191}$$

Note that this electric wave is one of the modes of the waves propagating in the wave guide. When considering the general situation, the total electric field is the superposition of each m and each n(for different wave number k_{mn}), that is:

$$\tilde{\mathbf{E}} = \sum_{\substack{m\in\mathbb{K}^+ \\ n\in\mathbb{N}}} \tilde{\mathbf{E}}_{mn},$$

therefore, the general solution to the electric field with the TM wave is:

$$\boxed{\tilde{\mathbf{E}} = \sum_{\substack{m\in\mathbb{K}^+ \\ n\in\mathbb{N}}} E_{0_{mn}} e^{im\phi} e^{i(k_{mn}z-\omega t)} \left[\hat{\mathbf{r}}\frac{iRk_{mn}}{s_{mn}} J_m'\left(\frac{s_{mn}}{R}s\right) - \hat{\boldsymbol{\phi}}\frac{k_{mn}mR^2}{s_{mn}^2 s} J_m\left(\frac{s_{mn}}{R}s\right) + \hat{\mathbf{z}} J_m\left(\frac{s_{mn}}{R}s\right)\right]} \tag{192}$$

We can use the same method to obtain the magnetic field in this TM wave. Conversely, when the wave becomes a TE wave, then $E_z = 0$, but we just do the same thing to obtain B_z, B_s and B_ϕ.

Chapter 9 End!